



Full project documentation

Subscriber Churn Diagnosis · data & BI portfolio project · synthetic, seed-generated data
Generated 2026-09-11

Voxa+ — Subscriber Churn Diagnosis

A data & BI portfolio project. **Voxa+** is a fictional SVOD (subscription video-on-demand) streaming service operating in Brazil, Mexico, and the United States. Monthly subscriber churn nearly doubled starting in month 34 of a 36-month window and has stayed elevated. Leadership needs a data-driven root-cause diagnosis before the next board meeting.

This repository simulates the full analytics stack end to end: messy source exports, a medallion ETL pipeline, a Kimball star schema, a code-generated Power BI semantic model and report, and a dual-track (SQL + Python) root-cause investigation.

All data is synthetic and generated from a single random seed. Any resemblance to real companies or people is coincidental.

Visual case study (PDF) — 7-page carousel summary.

Scope (locked in Phase 0)

Deliverable language	English
Data window	36 months; churn shift begins month 34
Markets / currencies	Brazil / BRL (home, month 1) · Mexico / MXN (month 10) · United States / USD (month 16)
Reporting currency	USD default, with a currency slicer (BRL / MXN / USD)
Data scale	~120k cumulative subscribers, ~70k active at window end; viewing at subscriber-day grain
North Star	Paid Active Subscribers · watch metric: Gross Monthly Churn %
Depth	Balanced (data engineering + analysis)
PBI environment	.pbip / PBIR, generated from code

Report pages (Phase 1)

- 1. Executive Summary — how bad is the spike, is it still going
- 2. Subscriber Retention & Cohorts — which cohorts / segments are driving it
- 3. Content & Engagement — did viewing behavior drop before cancellation
- 4. Pricing & Plans — did a price / plan change trigger it
- 5. Acquisition Quality & Channels — did subscriber quality shift
- 6. Customer Experience, Billing & App Quality — payment failures / app issues
- 7. Market Context — global or concentrated in one market

The diagnosis

The near-doubling of churn is **three overlapping factors**, none sufficient alone:

- 1. an **app "v3" connected-TV playback regression** (the acute trigger — CTV-heavy churn 4.1% → 8.4% while the rest went 4.1% → 5.6%);
- 2. the **Brazil price increase**, which did nothing to engaged subscribers and **+18 pp** to already-disengaged ones (the amplifier, Brazil-only);
- 3. a **months-old Mexico cohort-quality drag** (newer Mexican cohorts retain ~8 pp worse at month 3 — the pre-existing drain).

Payments were tested and ruled out. Deliverables (built by `data_analysis/deliverables/build_deliverables.py`): board read — `board_summary.pdf` / `.html` · full write-up with all ten findings — `deep_dive.pdf` / `.html` (source: `deep_dive_00_index.md`).

Repository layout

```
docs/          01-09 docs · 06_dax_measures & 03_data_dictionary auto-generated ·
               walkthrough.md (build log) ·
               build_docs_pdf.py → Voxa+_Documentation.pdf / Voxa+_Summary.pdf
etl/           config.py (one SEED) · gen/ world+content+economics · 01_generate_raw ·
               02_clean_stage · 03_build_curated · dq_checks · run_pipeline ·
               pbi_measures.py (113 DAX measures) · gen_data_dictionary / gen_dax_doc
data/          raw / staging / curated (Parquet + CSV mirror for compact tables) / quality
powerbi/       gen_semantic_model.py → VoxaChurnDiagnosis.SemanticModel (TMDL) ·
               gen_report_visuals.py + _drive_pbir.py + build_report.sh → 7-page report ·
               theme/theme.json · assets/ (brand PNGs)
assets/        brand.py (palette) · gen_logo.py (mark, wordmark, page backgrounds)
data_analysis/ run_analysis.py (10 questions × SQL + pandas, parity-checked) ·
               sql/ · results/ · parity/ · findings/F01-F10 · deliverables/
portfolio/     gallery/ (diagram + screenshot sources)
```

Reproduce

```
python -m pip install -r etl/requirements.txt
python etl/run_pipeline.py                # raw → staging → curated → dq_checks (~28 min)
cd powerbi && bash build_report.sh         # semantic model + 7-page report (Desktop closed)
python data_analysis/run_analysis.py      # the dual-track diagnosis
python docs/build_docs_pdf.py             # the documentation PDFs
```

Status

Phases 0–12 complete — pipeline, semantic model, report, dual-track diagnosis and deliverables all build clean. Build log in [docs/walkthrough.md](#).

Voxa+ — Business Context & KPI Framework

Document status: Phase 1 deliverable · authored at project kickoff, before any analysis. **Data generation date:** 2026-09-02 · **Analytics window:** 2023-09-01 → 2026-08-31 (36 full months). **Audience:** the Voxa+ leadership team and the board.

This document frames the problem and defines the metrics. It deliberately does **not** contain a conclusion about what caused the churn increase — that is the job of the investigation (Phase 10). Everything here was knowable *before* the diagnosis began.

1. The situation

Voxa+ is a subscription video-on-demand (SVOD) streaming service. For most of its history, monthly subscriber churn has run in a stable band. **Starting in month 34 of this window (June 2026), gross monthly churn nearly doubled and it has stayed elevated for the three months since.** Net subscriber additions have turned negative in the home market for the first time outside of a seasonal content gap.

Leadership has not been able to agree on a cause. Several things changed in the same broad period — pricing, the app, the content slate, the acquisition mix, the international footprint — and each function has a different theory. The board meets in five weeks and wants a single, evidence-backed diagnosis rather than seven competing narratives.

This project builds the data foundation and the analysis to deliver that diagnosis.

2. Company profile

Name	Voxa+ (stylised "Voxa+"; legal entity <i>Voxa Mídia S.A.</i>)
Headquarters	São Paulo, Brazil
Business	Direct-to-consumer SVOD streaming — films, series, and a growing slate of originals
Age	Operating in Brazil for ~6 years; the last 3 years (this window) cover the international expansion
Employees	~640 (not modelled below the level of allocated G&A)
Fiscal calendar	Calendar year; reporting is monthly
Scale at window end	~70k paid active subscribers, ~120k cumulative sign-ups across the window, ~US\$0.5M MRR (~US\$6M ARR)
Devices	Mobile (iOS/Android), web, and connected-TV apps (Smart TV, streaming sticks, consoles)

Positioning

Voxa+ is a **mid-priced general-entertainment** service — cheaper than the global majors, with a catalogue weighted toward Brazilian and Latin American content plus licensed international titles. Its edge is local originals and locally relevant pricing and payment methods (PIX and boleto in Brazil, OXXO and card in Mexico, card in the US).

3. The analytics window — why 36 months, and CY vs PY

Anchor	Value
Window length	36 complete months: month 1 = Sep 2023 , month 36 = Aug 2026
Current Year (CY)	months 25–36 → Sep 2025 – Aug 2026
Prior Year (PY)	months 13–24 → Sep 2024 – Aug 2025
Pre-shift baseline	months 1–33 (Sep 2023 – May 2026)
Churn shift	begins month 34 (Jun 2026) ; elevated in months 34, 35, 36

Why 36 months. It is the shortest window that contains the *entire* international expansion (both new markets launched inside it), gives a full CY-vs-PY comparison for the home market, and still leaves a long stable baseline before the shift so "normal" is well characterised. A shorter window would make the pre-shift band look like noise; a longer one would drag in a platform migration that predates the current data pipeline.

Comparability caveat. Because two markets launched mid-window, raw company-wide YoY mixes organic movement with expansion. Every time-series metric is therefore reported three ways: **Total**, **Comparable Base**, and **Expansion** (see §7).

4. Market & expansion timeline

Market	Currency	Enters window	Window-month	Notes
Brazil	BRL (R\$)	present from month 1	1 (Sep 2023)	Home market. The only market present for the full window → the Comparable Base.
Mexico	MXN (MX\$)	Jun 2024	10	Spanish-language localisation, OXXO cash payment, telco bundle partner from launch.
United States	USD (US\$)	Dec 2024	16	English UI, card-only, positioned against incumbents on price; smallest base, highest ARPU.

Launch dates are modelled explicitly. Like-for-like analysis uses Brazil (full history) as the Comparable Base; Mexico gains its own 12-month YoY from month 22 (Jun 2025), the US from month 28 (Dec 2025).

5. Product catalogue, pricing, and the margin structure

5.1 Plan line-up (per market, monthly billing)

Tier	Description	Brazil (R\$)	Mexico (MX\$)	US (US\$)
Voxa Básico	Ad-supported, 1080p, 1 stream, no downloads	18.90	89	6.99
Voxa Standard	Ad-free, 1080p, 2 streams, downloads	29.90	139	12.99
Voxa Premium	Ad-free, 4K HDR, 4 streams, downloads	44.90	219	17.99

- **Annual plans** are offered at all tiers for the price of 10 months (~17% discount), billed once and recognised monthly.
- A **7-day free trial** is available on Standard and Premium via direct sign-up (not via most partner/bundle channels).
- Prices shown are the line-up **as of window end**. The list price has not been static for the full 36 months — see §8 (events).

5.2 Revenue streams

Stream	~Share of revenue	Mechanics
Subscription	~86%	Recurring plan fees, reporting-currency converted; annual plans amortised straight-line over 12 months.
Advertising	~10%	CPM-based, served against Básico-tier viewing. Revenue ≈ ad impressions × fill rate × eCPM. Seasonal (upfront-style H2 strength).
Partner / bundle	~4%	Revenue-share and wholesale arrangements with telcos (carrier billing in BR/MX) and one retail bundle. Lower net price, near-zero acquisition cost.

5.3 Cost structure

Cost	Nature	Driver
Content — amortisation	Largest cost	Licensed titles amortised over licence term; originals over an accelerated useful-life curve. This is a CFO deep-dive thread (§9).
Content — cash spend	Cash, lumpy	Minimum guarantees and production milestones; diverges from amortisation month to month.
Streaming delivery (CDN)	Variable	~per GB delivered; scales with viewing hours and bitrate (4K costs more per hour).
Payment processing	Variable	% + fixed fee per attempt, varies sharply by method (card vs PIX vs boleto vs OXXO vs carrier billing). Also drives involuntary churn via failed renewals — CFO thread (§9).
Customer acquisition (marketing)	Semi-variable	Performance media + brand + partner bounties. Basis for CAC.
Customer support	Variable	Cost per contact; contact rate per 1,000 subs.
Cloud, platform & G&A	Largely fixed	Allocated flat per subscriber for margin views.

5.4 The core tension

For a subscription business the revenue-vs-profit tension is **growth vs. retention economics**:

- Cheaper tiers (Básico) and partner/bundle channels **add subscribers fast and cheap** but carry lower ARPU, lower engagement, and historically weaker retention.
- Premium tier and direct sign-ups **retain and monetise better** but cost more to acquire and are a smaller share of net adds.
- Content spend lifts engagement and retention, but with a lag, and it is the line most visible to the board when margin is under pressure.

A churn increase is therefore never "just a retention problem" — it moves LTV, which moves the allowable CAC, which moves the growth plan. The investigation has to quantify the **profit and LTV impact**, not only the rate.

6. Business model — financial definitions and the North Star

6.1 Subscriber & churn definitions (the spine of this project)

Term	Definition
Paid active subscriber	An account with an in-force paid subscription on the measurement date (trials excluded until they convert).
New paid subscriber	First paid billing in the period (trial conversion or direct paid start).
Reactivation	A previously-churned account starting a new paid subscription.
Gross subscriber churn (logo churn)	$\text{Paid subscriptions that ended in the month} \div \text{paid subscriptions active at the start of the month}$. Primary watch metric . Includes voluntary + involuntary.
Voluntary churn	Member-initiated cancellation (immediate or at period end).
Involuntary churn	Subscription ended because billing failed and the dunning sequence was exhausted.
Net subscriber adds	$\text{New} + \text{reactivations} - \text{gross churn}$.
Net MRR retention (NRR)	$\text{MRR from a starting cohort 12 months later (incl. upgrades, downgrades, churn, win-back)} \div \text{starting MRR}$. Secondary.
Cohort	All accounts with the same first-paid month.
Mn retention	Share of a cohort still paid-active n months after first paid billing.
Trial-to-paid conversion	$\text{Trials that convert to a first paid billing} \div \text{trials started}$.

6.2 Revenue & margin definitions

Term	Definition
MRR	$\Sigma (\text{active paid subscriptions} \times \text{normalised monthly plan price})$, reporting currency. Annual plans counted at 1/12.
ARPU	$\text{Subscription revenue} \div \text{average paid active subscribers in the period}$.
Total revenue	Subscription + advertising + partner/bundle.
Gross profit	Total revenue – content amortisation – CDN – payment processing – support.
Gross margin %	$\text{Gross profit} \div \text{total revenue}$.
Contribution margin	$\text{Gross profit} - \text{marketing (acquisition} + \text{retention/lifecycle)}$.
CM % (T12M)	$\text{Trailing-12-month contribution margin} \div \text{trailing-12-month revenue}$.
Blended CAC	$\text{Total marketing spend} \div \text{new paid subscribers (incl. trial conversions, excl. reactivations)}$. Scored lower-is-better .
LTV	$\text{ARPU} \times \text{gross margin \%} \times \text{expected lifetime}$, where $\text{expected lifetime} = 1 \div \text{monthly gross churn rate}$.
LTV : CAC	$\text{LTV} \div \text{blended CAC}$.
CAC payback (months)	$\text{Blended CAC} \div (\text{ARPU} \times \text{gross margin \%})$.

6.3 North Star and watch metric

- **North Star: Paid Active Subscribers (end of period)**. Everything the company does is in service of a growing, healthy paid base.
- **Watch metric: Gross Monthly Churn %**. The North Star cannot grow if the base leaks faster than acquisition can refill it. This is the metric that broke in month 34.

Supporting the North Star: NRR (value retention), engagement health (leading indicator of churn), and LTV:CAC (whether growth is economic).

7. Total vs Comparable Base vs Expansion

Cut	Definition	Use
Total	All markets, all subscribers.	Headline scale and the board number.
Comparable Base	Brazil only — the one market present for the entire window.	Like-for-like trend; strips expansion in/out of company-wide rates.
Expansion	Mexico + United States.	Isolates the contribution (and drag) of the new markets.

$\text{Expansion} = \text{Total} - \text{Comparable Base}$ for additive measures. Rates (churn %, ARPU, margin) are computed within each cut, never subtracted. Each market also carries its own `market_active_from` date so a per-market YoY only starts once 12 months of history exist.

An account-level flag `is_comparable_base` = (market = Brazil). A second flag `is_141_cohort` = (first-paid month ≤ month 24) supports 12-month cohort comparisons that are fully inside the pre-shift baseline.

8. Events in the window (neutral history)

These are things that happened. **None is asserted here to be a cause of the churn shift** — they are the candidate context the investigation will test, confirm, or rule out. Dates are approximate to the month.

~Month	Date	Event
6	Feb 2024	Cancellation flow redesigned (fewer steps, added a "pause subscription" option).
10	Jun 2024	Mexico launch — OXXO payment, telco bundle partner live from day one.
13	Sep 2024	Support platform migration (new ticketing tool; contact tagging taxonomy changes).
16	Dec 2024	US launch — card-only, price-led positioning.
18	Feb 2025	Ad-tier eCPM deal renegotiated upward; ad load per hour increased slightly.
22	Jun 2025	Two long-running licensed international series expire from the Brazil catalogue.
25	Sep 2025	Performance-marketing mix shifts toward lower-cost affiliate and paid-social inventory; a new telco bundle push begins in Brazil and Mexico.
28	Dec 2025	Annual-plan discount deepened for a holiday promo (echoes into renewals ~12 months later).
30–33	Feb–May 2026	Voxa+ app "v3" rollout — re-platformed player and navigation, shipped device-family by device-family (mobile first, connected-TV last).
31	Mar 2026	Brazil list-price increase on Standard and Premium; Básico held flat. Existing members migrated on their next renewal.
32	Apr 2026	Plan-lineup tweak: Básico stream quality and ad load adjusted; downloads removed from Básico.
34	Jun 2026	Churn steps up. <i>(This row is the symptom, not an event.)</i>

The investigation must weigh several of these against each other rather than seizing on the one nearest to month 34.

9. CFO-grade deep-dive threads

Two threads get modelled with extra rigour because they carry the profit and cash story:

9.1 Content economics

- **Amortisation vs cash spend** tracked separately; a monthly bridge between them.
- **Content cost per viewing hour** and **per retained subscriber**, by market.
- **Content supply cadence**: releases per month, days since the last "tentpole" title, and a share-of-catalogue-refreshed metric — the supply side of engagement.

9.2 Payments, dunning, and involuntary churn

- Payment **method mix** (card / PIX / boleto / OXXO / carrier billing) with method-level **authorisation rates**, **processing cost**, and **failure** → **recovery** funnels.
- **Dunning**: retry schedule, recovery rate by attempt, days-to-recovery, and the involuntary-churn tail when dunning is exhausted.
- Effect of the payment-method mix on **effective net price** and on **cash conversion** (carrier-billing settlement lag, boleto float).

Working-capital framing: subscription cash is collected close to service, so the classic cash-conversion-cycle story is thin; the cash levers here are **content cash timing** and **payment settlement lag**, reported as a simplified cash bridge rather than a full CCC.

10. Multi-currency handling

- Every monetary column is stored twice: `*_local` (transaction currency) and `*_usd` (converted at the month's average rate).
- A disconnected **Reporting Currency** table (BRL / MXN / USD) drives a slicer, with `FX Rate to Reporting Currency` (= 1 when reporting currency = USD) and a `Currency Symbol` measure (R\$ / MX\$ / US\$).
- Leaf monetary measures are wrapped `× [FX Rate to Reporting Currency]`; **USD is the default** reporting currency for the portfolio.
- FX paths over the window are seeded with plausible drift: **BRL ≈ 4.9–5.5 / USD**, **MXN ≈ 17–20 / USD**. Rates move gradually; no single month swings enough to explain a churn step by itself, but FX is available as a control.

11. Seasonality and patterns embedded in the data

Pattern	Where it shows
Dec–Jan Brazilian summer + year-end tentpoles	Acquisition and engagement peak; annual-plan sign-ups spike.
Carnaval (Feb)	Short engagement dip in Brazil, ~1–2 weeks.
Jun–Jul Festa Junina / winter	Engagement rises; a mid-year content push.
Content gaps (weeks between tentpoles)	Engagement troughs and a mild, <i>recurring, expected</i> churn uptick — the baseline seasonal churn rhythm.
Annual-plan renewal echoes	Sign-up peaks re-appear as renewal (and churn-decision) spikes ~12 months later.
Payment cadence	Boleto/PiX retries cluster at month start; carrier-billing subs bill and lapse on the partner's cycle.
Mexico	Semana Santa (Apr) and Día de Muertos (Nov/Dec) engagement bumps; December bundle promotions.
US	Post-holiday Q1 cancellations, a summer slump; short dip on Super Bowl weekend.
FX drift	Gradual BRL/MXN depreciation vs USD nudges USD-reported ARPU down even when local ARPU is flat.

The month-34 shift sits **on top of** this seasonal structure and is larger than any seasonal move in the baseline.

12. Stakeholder personas → report pages

Each persona brings one standing question. Those questions are the seven Phase-1 pages.

#	Persona	Standing question	Page
1	CEO / Board	How bad is the spike, and is it still getting worse?	Executive Summary
2	VP Retention & Lifecycle	Which cohorts, tenures, plans, markets, and sign-up sources are actually churning?	Subscriber Retention & Cohorts
3	VP Content & Programming	Did people stop watching <i>before</i> they cancelled? Is there a content-supply gap?	Content & Engagement
4	VP Pricing & Monetization	Did a price increase or plan change push people out — and how price-sensitive is each segment?	Pricing & Plans
5	VP Growth / Performance Marketing	Did the <i>quality</i> of who we acquire drop — worse channels, worse early retention?	Acquisition Quality & Channels
6	VP Customer Experience / CTO	Are payment failures or app problems forcing people out — involuntary churn, crashes, playback errors, support surge?	Customer Experience, Billing & App Quality
7	CFO / Head of Strategy	Is this global or concentrated in one market — and what is the revenue and LTV hit?	Market Context

No single page is expected to contain the whole answer; the diagnosis comes from reading them together.

13. KPI framework

Benchmark ranges are **directional**, based on publicly disclosed SVOD metrics and analyst commentary, adapted to a mid-priced Latin-America-led service. They set the targets; they are not claims about competitors.

13.1 Growth & retention

KPI	Definition	Benchmark range	Target
Paid Active Subscribers (EoP)	In-force paid accounts at period end	—	On plan curve (\$14)
Net Subscriber Adds / month	New + reactivations – gross churn	> 0 ex-seasonal	≥ plan
Gross Monthly Churn %	Ended paid subs ÷ start-of-month paid subs	3.0%–6.0%	≤ 4.0%
Voluntary Churn %	Member-initiated share of churn	2.2%–4.5%	≤ 3.0%
Involuntary Churn %	Dunning-exhausted share of churn	0.7%–1.6%	≤ 1.0%
Net MRR Retention (12M)	Cohort MRR after 12M ÷ starting MRR	92%–102%	≥ 99%
M1 Retention	Cohort active after first renewal	82%–90%	≥ 88%
M6 Retention	Cohort active at month 6	52%–70%	≥ 62%
M12 Retention	Cohort active at month 12	38%–55%	≥ 48%
Trial-to-Paid Conversion %	Converting trials ÷ trials started	45%–65%	≥ 55%
Reactivation Rate %	Win-back within 12M ÷ churned base	8%–16%	≥ 12%

13.2 Engagement (leading indicators of churn)

KPI	Definition	Benchmark range	Target
Monthly Active Rate %	$\text{Paid subs who streamed} \geq 1 \text{ min in the month} \div \text{paid subs}$	74%–88%	$\geq 85\%$
Viewing Hours / Active Sub / month	$\text{Streamed hours} \div \text{monthly active subs}$	40–70 h	45–65 h
% Subs Below Healthy Engagement	Paid subs with < 5 streamed hours in the month	10%–20%	$\leq 15\%$
Titles Started / Active Sub / month	Distinct titles with a play start	4–9	≥ 6
Days Since Last Tentpole Release	Recency of a major release, by market	< 45 d	$\leq 30 \text{ d}$
Content Releases / month	Titles added to catalogue	—	$\geq \text{plan}$

13.3 Monetisation & unit economics

KPI	Definition	Benchmark range	Target
ARPU (reporting currency)	$\text{Subscription revenue} \div \text{avg paid subs}$	BR ~US\$5.5–7 · MX ~US\$6–8 · US ~US\$11–14	$\geq \text{plan}$
MRR	Normalised recurring subscription value	—	$\geq \text{plan}$
Total Revenue YoY %	Total revenue vs prior year (per cut)	—	$\geq \text{plan}$
Gross Margin %	$\text{Gross profit} \div \text{revenue}$	28%–45%	$\geq 40\%$
Contribution Margin % (T12M)	$(\text{Gross profit} - \text{marketing}) \div \text{revenue}$	6%–18%	$\geq 12\%$
Blended CAC	$\text{Marketing} \div \text{new paid subs (lower is better)}$	US\$14–30 blended	$\leq \text{US\$22}$
CAC Payback (months)	$\text{CAC} \div (\text{ARPU} \times \text{gross margin \%})$	12–20	≤ 16
LTV : CAC	$\text{LTV} \div \text{blended CAC}$	≥ 3.0	≥ 3.5
% Revenue from Ads Tier	$\text{Advertising} \div \text{total revenue}$	6%–14%	context
% New Subs via Partner/Bundle	$\text{Bundle sign-ups} \div \text{new subs}$	10%–25%	context

13.4 Acquisition quality

KPI	Definition	Benchmark range	Target
New Subs by Channel	Direct / paid-search / paid-social / affiliate / partner-bundle / organic	—	mix on plan
Channel M1 Retention	M1 retention split by acquisition channel	direct highest, affiliate/bundle lowest	each $\geq$ its plan
Channel M3 Retention	M3 retention split by acquisition channel	—	each $\geq$ its plan
Trial Start Rate by Channel	$\text{Trials started} \div \text{sign-ups, by channel}$	—	context
Share of Adds from Low-Retention Channels	Affiliate + bundle share of new subs	20%–40%	$\leq 35\%$
Incentivised Sign-up Share	Sign-ups with a promo/discount code	25%–45%	$\leq 40\%$

13.5 Customer experience, billing & app quality

KPI	Definition	Benchmark range	Target
Payment Failure Rate %	$\text{Failed billing attempts} \div \text{attempts}$	4%–9%	$\leq 6\%$
Auth Rate by Method	$\text{Approved} \div \text{attempted, per payment method}$	card 88–95% · PIX 95–99% · boleto 70–85% · OXXO 65–80% · carrier 90–97%	per-method plan
Dunning Recovery Rate %	$\text{Recovered} \div \text{entered dunning}$	35%–55%	$\geq 45\%$
Days to Recovery	Mean days from failure to successful charge	2–7	≤ 5
App Crash-Free Session Rate %	$\text{Sessions with no fatal crash} \div \text{sessions}$	99.2%–99.8%	$\geq 99.5\%$
Playback Error Rate %	$\text{Streams with a fatal playback error} \div \text{streams}$	0.3%–1.2%	$\leq 0.7\%$
Video Start Failure %	$\text{Play attempts that never start} \div \text{attempts}$	0.5%–2.0%	$\leq 1.2\%$
Support Contacts / 1,000 Subs	$\text{Tickets} \div (\text{subs} / 1,000) \text{ per month}$	40–90	≤ 70
CSAT	Mean post-contact satisfaction (1–5)	3.9–4.5	≥ 4.2

13.6 Market context

KPI	Definition	Use
Churn % by Market	Gross monthly churn split BR / MX / US	Is the shift global or concentrated?
Net Adds by Market	Monthly net adds per market	Where the base is actually shrinking
Revenue & MRR by Market	Reporting-currency, with FX effect isolated	Size of the financial hit
Engagement & App Quality by Market	\$13.2 / \$13.5 metrics per market	Whether operational health differs by market
Cumulative Subscribers Lost vs Plan	(Plan base – actual base), by market, since month 34	The board-level scoreboard of the gap

14. Targets and plan

- The plan is built on the **same basis as the actuals** (net of involuntary churn where the actual measure is; local-currency plan then FX-converted, not the reverse).
- **Plan subscriber base**: $PY\ actual \times (1 + \text{planned growth})$ per market per month, with planned growth set *before* the shift (Brazil mid-single-digit YoY, Mexico and US ramping from their launch curves).
- **Plan churn**: flat at the \$13.1 targets (gross $\leq 4.0\%$, involuntary $\leq 1.0\%$).
- **Plan ARPU / MRR**: plan base $\times$ plan ARPU (local), then converted.
- **Plan marketing / CAC**: spend to hit plan net adds at $CAC \leq US\$22$ blended.
- Plan is monthly for the full window; the CY plan (months 25–36) is the one shown against actuals on the report. "vs Target %" measures exist per KPI, with CAC and churn scored **lower-is-better**.

15. Out of scope (explicit)

- Title-level recommendation / personalisation algorithm analysis.
- Ad-tech yield and programmatic optimisation beyond blended eCPM and fill rate.
- Device- or OS-level quality-of-experience beyond crash rate, playback-error rate, and video-start-failure rate.
- Tax, transfer pricing, and statutory / IFRS reporting detail.
- Headcount, payroll, and opex below allocated G&A.
- A churn-**prediction** model. This project is a **diagnosis**, not a forecast.
- External data: competitor pricing, market share, third-party panel data. The "Market Context" page is internal markets (BR/MX/US), not the competitive market.
- Real personal data. All subscribers, payments, and viewing events are synthetic and seed-generated.

16. Resolved-assumptions log

#	Decision	Resolution
A1	Window length & anchor	36 months, Sep 2023 – Aug 2026; data-gen date 2026-09-02.
A2	Market launch months	Brazil = pre-window; Mexico = month 10 (Jun 2024); US = month 16 (Dec 2024).
A3	Churn-shift timing	Steps up at month 34 (Jun 2026); elevated for months 34–36.
A4	Primary churn metric	Gross monthly logo churn; voluntary/involuntary split reported alongside.
A5	North Star	Paid Active Subscribers (EoP); watch metric = gross monthly churn %.
A6	Plan lineup	Three tiers (Básico ad-supported / Standard / Premium), monthly + annual, per-market local pricing.
A7	Trial	7-day free trial on Standard/Premium via direct sign-up only.
A8	Comparable Base	Brazil only (sole full-window market); Expansion = Mexico + US.
A9	Reporting currency	USD default; slicer for BRL/MXN/USD; *_local + *_usd on every money column.
A10	FX method	Monthly average rates, seeded gradual drift; no single-month FX shock.
A11	Content accounting	Amortisation and cash spend modelled separately, with a monthly bridge.
A12	Payment scope	Method mix, auth rates, dunning funnel, involuntary-churn tail modelled; full CCC not.
A13	Fiscal calendar	Calendar year; monthly reporting.
A14	Benchmark ranges	Directional, from public SVOD disclosures/analyst ranges; used only to set targets.
A15	Root cause	Deliberately not decided in this document. Determined during ETL (Phase 3), surfaced only through the data and the Phase 10 investigation.

Voxa+ — Data Architecture

Document status: Phase 2 deliverable. **Depends on:** 01_business_context_kpis.md . **Feeds:** Phase 3 (ETL build), Phase 4 (data dictionary / model docs), Phase 6 (semantic model).

This document specifies how synthetic source data is generated, cleaned, and conformed into a star schema. It is a build spec: every table, every layer, every data-quality issue that Phase 3 must inject and then handle.

1. Principles — medallion, seeded, idempotent

Three layers on disk. Data only ever flows downward.

Layer	Folder	Contract
Raw	data/raw/<source>/	Byte-faithful simulated exports — messy, locale-specific, duplicated, encoded however the fake source system would encode them. Never edited after generation. Regenerated only when SEED or config.py changes.
Staging	data/staging/	Exactly one typed table per raw entity: stg-<source>_<entity>.parquet . Parsing, type coercion, encoding repair, de-duplication, key and category normalisation. No cross-source business logic, no metric definitions. Every fix is counted into a DQ fragment.
Curated	data/curated/	Conformed Kimball star: dim_* / fact_* as Parquet and a CSV mirror. Surrogate keys, SCD handling, the movement-ledger-derived snapshot, annual-plan monthly recognition, content cost per view-hour, USD on every money column, is_comparable_base .
Quality	data/quality/	DQ fragments from staging + curated, assembled into dq_report.md ; hard-assertion results from dq_checks.py .

Seeded. A single SEED in etl/config.py drives every random draw. Same seed → byte-identical raw → identical curated.

Idempotent. Each script wipes and rebuilds its own layer. run_pipeline.py runs 01 → 02 → 03 → dq_checks end to end; dq_checks.py fails the build (non-zero exit) on any violated assertion.

Layered. Staging may read only raw. Curated may read only staging. Nothing reads upward. The semantic model (Phase 6) reads only data/curated/*.parquet .

2. Tech stack & run instructions

Concern	Choice
Language	Python 3.11+
Core libs	pandas , numpy , pyarrow (Parquet), python-dateutil , openpyxl (write messy .xlsx), Faker (names, emails, addresses), ftfy + unicodedata (encoding repair in staging), duckdb (Phase 10 analysis + optional staging SQL)
Storage	Flat files. Raw = .csv / .jsonl / .xlsx / .csv.gz / .parquet . Staging & curated = .parquet (+ curated .csv mirror). No database server.
Orchestration	Plain scripts, run in order by run_pipeline.py . No Airflow / dbt.
Determinism	One SEED ; all generators derive sub-seeds from it by name.
Windows / OneDrive	Writes go to a temp path in the same folder, then atomic os.replace . A retry-on-lock wrapper (utils.write_parquet , ~4 attempts, backoff) absorbs the intermittent WinError 5/32 locks OneDrive causes.

```
# from repo root
python -m pip install -r etl/requirements.txt
python etl/run_pipeline.py           # full build: raw → staging → curated → dq_checks
python etl/run_pipeline.py --from 02 # rebuild staging + curated only
python etl/gen_data_dictionary.py    # regenerate docs/03_data_dictionary.md from disk
```

Run order inside run_pipeline.py :

- 01_generate_raw.py — writes data/raw/**
- 02_clean_stage.py — writes data/staging/stg_*.parquet + data/quality/dq_fragments/stg_*.json
- 03_build_curated.py — writes data/curated/{dim,fact}*_.{parquet,csv} + curated DQ fragments
- dq_checks.py — hard assertions; assembles data/quality/dq_report.md ; exits non-zero on failure

3. Naming conventions

Thing	Rule	Example
Raw file	<source>__<entity>__<period-or-grain>.<ext>	pagstream__transactions__2024-06.csv
Staging table	stg_<source>_<entity>	stg_pagstream_transactions
Dimension	dim_<noun>	dim_subscriber
Fact	fact_<process>_<grain>	fact_billing_attempt
Surrogate key	<entity>_key — integer, curated only	subscriber_key
Business key	<entity>_id — preserved from source	subscriber_id
Date	<role>_date (date) · <role>_date_key (int YYYYMMDD)	signup_date, signup_date_key
Timestamp	<role>_ts_utc (tz-aware UTC)	event_ts_utc
Money	<measure>_local and <measure>_usd	mrr_local, mrr_usd
Boolean	is_<x> / has_<x>	is_active, has_downloads
Columns	snake_case throughout curated (staging harmonises source casing)	

Timestamps are normalised to **UTC** in staging. Curated derives a **market-local calendar day** (`America/Sao_Paulo` , `America/Mexico_City` , `America/New_York`) and the `*_date_key` from that local day, so month-boundary events land in the right reporting month per market.

4. Source-system inventory

Thirteen simulated systems. "Export style" is what makes each one messy.

#	System (fictional)	Owns	Raw entities	Export style / quirks
1	Ketch Billing & Subscriptions	Subscriptions, invoices, plan catalogue	subscriptions_snapshot , invoices , plan_catalog	Daily full-dump snapshot CSV (same rows re-exported every day → dupes); invoices split into monthly files with overlapping date ranges ; plan_catalog.xlsx with Excel serial effective dates and price as text "R\$ 29,90" .
2	PagStream Payment Gateway	Billing attempts, auth/capture/refund, settlement	transactions , settlement	One row per attempt; a semi-structured meta JSON blob column; amount is gross for approved rows but net-of-fee for settled rows (mislabelled); refunds as negative amounts in some months, type='refund' positive rows in others; timestamps in UTC with Z .
3	VoxalD Identity	Accounts, email, market, device registrations, consent	accounts , device_registrations	accounts.csv UTF-8 with BOM ; market as free text (BR / Brasil / Brazil); GDPR-deleted accounts absent though still referenced downstream; test accounts not flagged.
4	Entitlement Service	Subscription lifecycle movement ledger	entitlement_events	Newline-delimited JSON (.jsonl); event types trial_start , trial_convert , paid_start , plan_change , pause , resume , cancel_request , churn , reactivate ; at-least-once delivery → duplicate event_id s; the source of truth the curated snapshot is rebuilt from.
5	Bonsai CRM	Customer attributes, lifecycle stage, win-back campaign membership	crm_contacts , winback_membership	Latin-1 encoded names/titles → mojibake when read as UTF-8; lifecycle stage free text; birth_year occasionally 0 or 1900 ; joined late so early cohorts have blank CRM rows.
6	Trackpad App Analytics	App sessions, screens, crashes, app version, device	app_events (daily)	app_events__YYYY-MM-DD.csv.gz , event-sampled (~ not every session), at-least-once → duplicate event_id ; device_family values unnormalised (SmartTV / Smart TV / CTV); app_version string; UTC timestamps.
7	PlayLog Streaming Telemetry	Play starts, watch time, bitrate, playback errors, rebuffering, CDN bytes	playlog (daily)	playlog__YYYY-MM-DD.parquet ; watch time in seconds before ~month 20, milliseconds after an SDK update (unit break); CDN bytes in MB from one edge provider, GB from another ; a batch of negative watch_seconds from clock skew; content_id s that don't exist in Reelbase.
8	Reelbase Content Catalogue & Rights	Titles, genre, language, originals flag, licence windows	titles , licence_windows	reelbase_titles.xlsx , multiple sheets; accented PT/ES titles with encoding damage ; release_date mixes DD/MM/YYYY and Excel serials; some titles missing a genre; is_original as "Y"/"N"/""/1/0 .
9	Content Finance	Amortisation schedules, minimum guarantees, cash milestones	amort_schedule , cash_milestones	.xlsx , one sheet per content vintage; amounts as text with parentheses negatives "(120.000,00)" ; month columns as Excel serials ; MG in local currency, no currency column (implied by sheet name).
10	AdBridge Marketing Spend	Spend, impressions, clicks, attributed sign-ups by channel/campaign	spend_weekly , attribution	spend_weekly.csv ; channel names not normalised (Paid Social / paid_social / FB/IG / Meta); for the Mexico platform the spend_usd header actually holds MXN ; one column mislabelled — clicks holds impressions for a partner-network row; a test campaign with spend=0 , 10k sign-ups.
11	Voxa Ads SSP	Ad impressions served vs Básico viewing, fill rate, eCPM, ad revenue	ssp_daily_revenue	ssp_daily_revenue.csv ; date as MM/DD/YYYY (US-formatted export); revenue in USD; occasional duplicated day rows from re-runs.
12	Helpline → Helpdesk Support	Support tickets, reason, resolution time, CSAT	helpline_v1_tickets (≤ month 13), helpdesk_v2_tickets (≥ month 13)	Schema break at the mid-window migration (event \$8, doc 01): different column names, different reason taxonomy, different id format; ~some tickets duplicated across the cutover ; CSAT 1–5 in v1, 0–10 in v2.
13	Ledger (Finance GL) + FX	Monthly P&L actuals by line, allocations, FX rates	gl_trial_balance (monthly), fx_rates	gl_trial_balance__YYYY-MM.csv ; account names free text; fx_rates.csv monthly average and end-of-period rates for BRL/USD and MXN/USD, a few months missing (forward-fill in staging).

Two of these correspond to **neutral history events** in doc 01 \$8: the support migration (#12) and, indirectly, the marketing-mix shift (#10). Neither is asserted to be a cause of anything.

5. Raw layer — what 01_generate_raw.py emits

Grouped by simulated system. Volumes are approximate at SEED default (balanced scale).

Raw file(s)	Grain	~Rows	Key messy traits injected
ketch__subscriptions_snapshot__YYYY-MM-DD.csv (daily)	account × snapshot day	~9M total (dupe-heavy)	full re-dump daily; drifts from the ledger ; plan name casing noise; country free text
ketch__invoices__YYYY-MM.csv	invoice line	~1.6M	overlapping monthly ranges → cross-file dupes; money as "R\$ 1.234,56" ; annual plans as a single line
ketch__plan_catalog.xlsx	plan × market × effective period	~60	Excel-serial dates; price as text; retired plans with null successor
pagstream__transactions__YYYY-MM.csv	billing attempt	~2.1M	meta JSON blob; amount gross-vs-net mislabel; negative refunds; attempt_number, dunning flags
pagstream__settlement__YYYY-MM.csv	settlement batch line	~40k	settlement lag; fee columns; carrier-billing batches lag one month
voxaids__accounts.csv	account	~120k	UTF-8 BOM; free-text market; GDPR-deleted omitted; unflagged test accounts
voxaids__device_registrations.csv	account × device	~260k	unnormalised device_family ; duplicate registrations
entitlement__events.jsonl	lifecycle event	~950k	duplicate event_id ; out-of-order timestamps; the movement source of truth
bonsai__crm_contacts.csv	account	~110k	Latin-1 mojibake; free-text lifecycle stage; birth_year 0/1900; missing early cohorts
bonsai__winback_membership.csv	account × campaign	~70k	campaign name noise; overlapping memberships
trackpad__app_events__YYYY-MM-DD.csv.gz (daily)	app event (sampled)	~11M	duplicate event_id ; device/version noise; UTC; crash events interleaved
playlog__YYYY-MM-DD.parquet (daily)	play/watch beacon	~14M	seconds→ms unit break ~month 20; MB/GB byte-unit split; negative watch_seconds batch; orphan content_id
reelbase__titles.xlsx	title	~3.5k	encoding damage; mixed date formats; missing genre; is_original mixed types
reelbase__licence_windows.csv	title × market × window	~9k	licence start/end; some windows expiring mid-window (\$8 event)
contentfin__amort_schedule.xlsx	title × month	~40k	parentheses negatives; Excel-serial month columns; implied currency
contentfin__cash_milestones.xlsx	title × milestone	~1.2k	lumpy cash dates; MG local currency
adbridge__spend_weekly.csv	week × market × channel × campaign	~30k	channel-name noise; MXN-in-USD-column; clicks /impressions mislabel; zero-spend test campaign
adbridge__attribution.csv	sign-up × touch	~140k	last-touch only; some sign-ups unattributed (→ "unknown" channel)
voxaads__ssp_daily_revenue.csv	market × day	~3.2k	MM/DD/YYYY dates; duplicate day rows
helpline__v1_tickets.csv	ticket (≤ m13)	~55k	v1 schema; CSAT 1–5
helpdesk__v2_tickets.csv	ticket (≥ m13)	~120k	v2 schema; CSAT 0–10; cutover dupes
ledger__gl_trial_balance__YYYY-MM.csv	month × GL account	~4k	free-text account names; allocation lines
ledger__fx_rates.csv	month × currency pair	~150	avg + eop rate; a few months missing

6. Staging layer — 02_clean_stage.py

One `stg_*` table per raw entity above. Staging responsibilities, and the DQ counter each raises:

Fix class	What happens	DQ counter(s)
Date parsing	Detect and parse YYYY-MM-DD, DD/MM/YYYY, MM/DD/YYYY, DD-MON-YY, ISO-8601 with TZ, and Excel serials (epoch 1899-12-30). Ambiguous 01/02/2025 resolved by source locale rule.	dates_parsed, excel_serials_converted, dates_unparseable
Money parsing	Strip currency symbols and thousands separators by locale; "(x)" → -x; "R\$ 1.234,56" → 1234.56; blank → null (never 0).	money_strings_parsed, money_negatives_from_parens
Encoding repair	ftfy.fix_text + targeted Latin-1→UTF-8 re-decode for Reelbase / CRM; strip BOM.	mojibake_repaired, bom_stripped
De-duplication	Drop exact-duplicate event_id (entitlement, app, SSP); collapse daily full-dump snapshots to one row per account per day; dedupe overlapping invoice files on invoice_id.	dupe_events_dropped, dupe_invoice_lines_dropped, snapshot_rows_collapsed
Unit normalisation	PlayLog watch time → seconds (detect ms era by magnitude + date); CDN bytes → GB ; CSAT → 0–1 score (v1 /5, v2 /10).	watchtime_unit_fixed, bytes_unit_fixed, csat_rescaled
Category harmonisation	Map channel, plan-name, device-family, market free text to controlled vocabularies (lookup tables in etl/vocab/). Unknowns → explicit "unknown" + logged.	categories_normalised, category_unknowns
Mislabelled columns	AdBridge: move partner-network clicks → impressions; convert MX platform spend_usd → spend_local (MXN) and recompute USD. PagStream: reconstruct gross amount for settled rows from fee columns.	mislabeled_columns_corrected
Key normalisation	Trim/upper business keys; cast to string; null-out placeholder keys ("0", "", "N/A").	null_keys_flagged
Schema unification	Map Helpline v1 + Helpdesk v2 to one stg_support_tickets schema and one reason taxonomy (crosswalk in etl/vocab/support_reason_crosswalk.csv); tag source_version.	support_rows_v1, support_rows_v2, cutover_dupes_dropped
Type coercion	Enforce a declared dtype per column; coercion failures → null + counter.	type_coercion_failures
Timezone	All timestamps → tz-aware UTC.	timestamps_localized

Staging does **not** join across sources (except attaching a normalised key), does **not** compute MRR / churn / margin, and does **not** drop rows for referential problems — that is curated's job, with a full accounting.

7. Curated layer — 03_build_curated.py

Kimball star. dim_date is built first and spans **2023-01-01 → 2027-12-31** (well beyond the fact window, per the guardrail about a stray "(blank)" member when cash / renewals settle past the end).

7.1 Dimensions

Table	Grain	Notable attributes
dim_date	calendar day	date_key , y/q/m/w, month-name, is_month_end , fiscal cols, is_holiday_br/mx/us , season_tag , hidden sort keys
dim_market	market	market_id (BR/MX/US), currency_code , market_active_from_date , timezone
dim_subscriber	account	subscriber_key , subscriber_id , market_key , signup_date_key , cohort_month , acquisition_channel_key , acquisition_campaign_key , first_plan_key , current_plan_key , is_annual_ever , age_band , is_comparable_base (market = BR), is_l4l_cohort (cohort_month ≤ 2025-08), is_test_account (recovered flag), crm_lifecycle_stage
dim_plan	plan × market	plan_key , tier (Básico/Standard/Premium), billing_period (monthly/annual), has_ads , max_streams , max_resolution , has_downloads
dim_price_history	plan × market × effective period	list_price_local , list_price_usd , effective_from_date_key , effective_to_date_key — the price-change events live here
dim_channel	acquisition channel	channel (direct / paid_search / paid_social / affiliate / partner_bundle / organic / unknown), channel_group , is_incentivised_default
dim_campaign	marketing campaign	campaign_key , channel_key , market_key , start/end , objective
dim_payment_method	method	method (card / pix / boleto / oxoxo / carrier_billing), is_prepaid , settlement_lag_days_typical
dim_content	title	content_key , title (repaired), content_type , primary_genre , language , is_original , release_date_key , runtime_min
dim_content_release	title × market	release_date_key , is_tentpole , had_marketing_push
dim_device	device family	device_family (mobile_ios / mobile_android / web / smart_tv / streaming_stick / console), device_class (mobile / web / ctv)
dim_support_reason	reason	unified taxonomy: reason_category , reason_group (billing / app_tech / content / account / other)
dim_app_version	app version	app_version , release_date_key , device_class , is_major
dim_reporting_currency (disconnected — built in Phase 6)	currency	drives the currency slicer

7.2 Facts

Table	Grain	Measures / columns	Build notes
fact_subscription_event	one lifecycle event	event_type, event_ts_utc, event_date_key, from_plan_key, to_plan_key, reason_code	Deduped, ordered movement ledger. The source of truth.
fact_subscription_month	subscriber × month	is_active_eom, status (active/paused/churned), plan_key, tenure_months, mrr_local, mrr_usd, is_new, is_reactivation, is_voluntary_churn, is_involuntary_churn, is_paused	Derived from fact_subscription_event , not from the Ketch snapshot. Snapshot-vs-ledger drift is measured and written to the DQ report. Annual plans recognised at list_price / 12 per active month.
fact_billing_attempt	one billing attempt	amount_local/usd, fee_local/usd, net_local/usd, method_key, status (approved/failed/refunded/chargeback), failure_reason, attempt_number, is_dunning, is_recovered, days_to_recovery	Refund/chargeback signs normalised; fee reconstructed; linked to subscription + method.
fact_viewing_daily	subscriber × local date	streamed_minutes, play_starts, distinct_titles, primary_device_family_key, playback_errors, video_start_failures, rebuffer_ratio, bitrate_tier, cdn_gb	Aggregated from stg_playlog; watch time in seconds→minutes; orphan content_id rows kept with content_key = -1 (unknown) and counted. Subscriber-level QoE preserved so engagement/quality can be tied to individual churn.
fact_viewing_title_month	subscriber × title × month	streamed_minutes, completed	Slim rollup for content-level engagement cuts (kept small by monthly grain).
fact_app_performance_daily	market × date × device_family × app_version	sessions, crashes, crash_free_rate, anr_count	Aggregated app-health fact for the CX page; subscriber-level crash counts also folded into fact_viewing_daily.
fact_engagement_month	subscriber × month	active_days, streamed_hours, distinct_titles, pct_days_ctv, below_healthy_engagement (bool, < 5h), engagement_decile	Curated rollup — the leading-indicator table for cohort/retention analysis.
fact_marketing_spend	week × market × channel × campaign	spend_local/usd, impressions, clicks, signups_attributed	Post-normalisation of AdBridge; test campaign quarantined with is_excluded = true.
fact_ad_revenue	market × date	impressions_served, fill_rate, ecpm_usd, ad_revenue_usd	From SSP; joined to Básico viewing for sanity.
fact_content_cost	title × month	amort_usd, cash_spend_usd, mg_local/usd	Parentheses negatives fixed; Excel-serial months resolved; content cost per view-hour derived by joining to fact_viewing_title_month.
fact_support_ticket	one ticket	created_date_key, contact_channel, reason_category_key, resolution_hours, csat_score (0–1), is_reopened, source_version	Unified v1/v2; cutover dupes removed.
fact_fx_rate	month × currency pair	avg_rate, eop_rate	Missing months forward-filled in staging; = 1 for USD/USD.
fact_finance_month	month × market × pnl_line	amount_local/usd	From GL; conformed to the P&L line list; ties to the operational facts within tolerance (a DQ check).
fact_targets_month	month × market × kpi	target_value	Plan from doc 01 §14; built local then FX-converted.

7.3 Cross-cutting curated rules

- **Snapshot from ledger.** fact_subscription_month is folded forward from fact_subscription_event. The Ketch snapshot is used **only** to measure drift (rows where snapshot status ≠ ledger-derived status), reported as a DQ metric, never as input.
- **Annual-plan explode.** One annual invoice → 12 monthly MRR recognitions; cash vs recognition tracked separately.
- **Content cost per view-hour** = title-month amort_usd ÷ title-month streamed hours (moving denominator), with a floor to avoid divide-by-near-zero on unreleased titles — the SVOD analogue of moving-average COGS.
- **USD everywhere.** Every _local_ money column gets a _usd_ sibling at the month's avg_rate.
- **Referential integrity.** Facts referencing a missing dimension key get a sentinel key (-1 unknown / -2 late-provisioned) and a counted DQ row; they are **not** dropped.
- **is_comparable_base** stamped on dim_subscriber and propagated to facts via the subscriber key.
- **Local calendar day** per market drives every *_date_key_ on event-grain facts.

8. Data-quality issue catalogue (the injection list)

Every item is injected by `01_generate_raw.py` , handled by `02 / 03` , counted into a DQ fragment, and (where it is an invariant) asserted in `dq_checks.py` .

#	Issue	Injected in	Handled in	Hard check
Q01	Multiple date formats + Excel serials	plan_catalog, reelbase, contentfin, ssp	staging date parser	<code>dates_unparseable = 0</code>
Q02	Money as locale strings, parentheses negatives, symbol-in-value	invoices, contentfin, adbridge	staging money parser	no non-null money column is of object dtype
Q03	Encoding damage (Latin-1 mojibake, BOM)	reelbase, bonsai, voxaid	<code>ftfy</code> + re-decode	no <code>Ã / Â</code> sequences remain in <code>dim_content.title</code>
Q04	Duplicate rows — event at-least-once, overlapping invoice files, daily full-dump snapshot	entitlement, trackpad, ssp, ketch	staging de-dup	<code>fact_subscription_event</code> unique on <code>event_id</code> ; invoices unique on <code>invoice_id</code>
Q05	Master rows missing — orphan subscriber_id / content_id (GDPR-deleted, late-provisioned, unlisted titles)	voxaid omissions, playlog	curated sentinel keys + count	every fact FK resolves (to a real or sentinel key)
Q06	Mislabelled column — <code>clicks</code> holds impressions; <code>spend_usd</code> holds MXN; <code>amount</code> gross-vs-net	adbridge, pagstream	staging column correction	marketing <code>clicks ≤ impressions</code> for all rows
Q07	Snapshot-vs-ledger drift	ketch snapshot vs entitlement ledger	curated builds snapshot from ledger; drift measured	drift rows reported; curated status = ledger status by construction
Q08	Sign / negative errors — negative refunds, negative watch_seconds, negative days_to_recovery	pagstream, playlog	staging sign normalisation	<code>fact_viewing_daily.streamed_minutes ≥ 0</code> ; refunds stored with consistent sign
Q09	Unit inconsistency — watch seconds→ms, CDN MB vs GB, CSAT /5 vs /10	playlog SDK era, edge providers, support migration	staging unit normalisation	watch minutes per sub-day ≤ 1440; <code>csat_score ∈ [0,1]</code>
Q10	Categorical noise — channel / plan / market / device free text	adbridge, ketch, voxaid, trackpad	staging vocab mapping	all <code>dim_*</code> category columns ∈ controlled vocab
Q11	Schema break — Helpline v1 → Helpdesk v2	support source	staging schema unification + crosswalk	<code>fact_support_ticket</code> single schema; taxonomy ∈ <code>dim_support_reason</code>
Q12	Late / partial / restating files — partner settlement lag, incomplete final month	pagstream settlement, partner recon	staging forward-fill + <code>is_partial_period</code> flag on <code>dim_date</code>	last-month completeness flagged, not silently averaged
Q13	Null keys — guest billing rows, retired plan_id	pagstream, ketch	staging null-flag; curated sentinel	null-key count reported; no null surrogate keys in facts
Q14	Outliers — 30h+/day viewing, US\$0/10k-signup test campaign, annual invoice as one line	playlog, adbridge, invoices	curated caps / quarantine / annual explode	<code>fact_marketing_spend</code> excludes quarantined campaign from KPIs
Q15	Timezone / month-boundary bleed	UTC events vs local billing	curated local-day derivation	monthly viewing totals reconcile to daily within tolerance

`dq_checks.py` also asserts the **accounting / roll-up identities**: subscription-month base rolls forward (`start + new + reactivation - churn = end`), MRR ties to $\Sigma(\text{active} \times \text{price}/12)$ within tolerance, `fact_finance_month` subscription revenue ≈ curated MRR within tolerance, cohort retention ≤ 100% and monotonic non-increasing, and every KPI in doc 01 §13 lands inside its stated benchmark band for the **pre-shift** baseline (the shift months are allowed to break the band — that is the point).

9. Architecture diagram

```
flowchart TD
    subgraph SRC["13 simulated source systems"]
        S1["Ketch Billing & Subscriptions"]
        S2["PagStream Payment Gateway"]
        S3["VoxaID Identity"]
        S4["Entitlement Service (movement ledger)"]
        S5["Bonsai CRM"]
        S6["Trackpad App Analytics"]
        S7["PlayLog Streaming Telemetry"]
        S8["Reelbase Content Catalogue & Rights"]
        S9["Content Finance"]
        S10["AdBridge Marketing Spend"]
        S11["Voxa Ads SSP"]
        S12["Helpline / Helpdesk Support"]
        S13["Ledger GL + FX"]
    end

    SRC -->| "01_generate_raw.py (seeded, messy exports)" | RAW["data/raw/** CSV · JSONL · XLSX · CSV.GZ · Parquet"]
    RAW -->| "02_clean_stage.py parse · repair · de-dup · normalise + DQ fragments" | STG["data/staging/<br/>stg_*.parquet (typ"]
    STG -->| "03_build_curated.py conform · surrogate keys · snapshot-from-ledger<br/>· annual explode · USD everywhere" | CUR["dat"]
    STG --> DQF["data/quality/dq_fragments/"]
    CUR --> DQF
    DQF -->| "dq_checks.py hard assertions + identities" | DQR["data/quality/dq_report.md<br/>(build fails on violation)"]

    CUR -->| "Phase 6<br/>gen_semantic_model.py" | SM["Power BI .SemanticModel<br/>TMDL + Parquet import partitions"]
    SM -->| "Phase 8<br/>gen_report_visuals.py + build_report.sh" | RPT["Power BI .Report<br/>7 pages"]
    CUR -->| "Phase 10<br/>DuckDB (SQL) + pandas (Python)" | ANA["data_analysis/<br/>dual-track investigation"]
```

10. Open items carried into Phase 3

#	Item	Resolution plan
B1	Exact churn-shift mechanism (the 2–3 contributing factors)	Decided in Phase 3 , encoded in <code>config.py</code> as tunable effect sizes; kept out of doc 01.
B2	Final row volumes vs the ~10–15M viewing target	Calibrate <code>SEED</code> -default population and sampling in the first regeneration pass.
B3	Benchmark-band calibration (AOV/ARPU, margin, churn, CAC, retention, delivery, on-time)	Several regeneration passes until the pre-shift baseline sits mid-band for every doc 01 §13 KPI.
B4	Holiday/season calendars for BR/MX/US	Static lookup in <code>etl/vocab/holidays.csv</code> .
B5	FX path shape	Seeded random walk with drift, clamped to the doc 01 §10 ranges; no single-month shock.

End of Phase 2 deliverable.

Voxa+ — Data Dictionary (curated layer)

Auto-generated from `data/curated/` on 2026-09-02 by `etl/gen_data_dictionary.py`. Do not edit by hand.

All monetary columns are stored `*_local` (transaction currency) and `*_usd` (month-average FX). Surrogate keys are `*_key` (integer); business keys keep their source form as `*_id`. Sentinel keys: `-1` unknown, `0` unresolved.

dim_app_version

one row per observed app version string · 15 rows

column	dtype	sample values
app_version_key	int64	1, 2, 3
app_version	str	2.0.4, 2.1.4, 2.2.4
major	int64	2, 3
is_v3	bool	False, True
device_class_hint	str	all

dim_channel

one row per acquisition channel · 7 rows

column	dtype	sample values
channel_key	int64	1, 2, 3
channel	str	direct, organic, paid_search
channel_group	str	owned, performance, partner
is_low_quality	bool	False, True

dim_content

one row per catalogue title · 3,500 rows

column	dtype	sample values
content_key	int64	1, 2, 3
content_id	str	T50000, T50001, T50002
title	str	Días de Sol 2, Marea Alta, Maré 5
content_type	str	series, movie
primary_genre	str	Comedy, Crime, Unknown
language	str	es, pt, en
is_original	bool	False, True
runtime_min	int64	501, 124, 390
release_date_key	int64	20231201, 20220101, 20220501

dim_date

one row per calendar day · 1,826 rows

column	dtype	sample values
date	datetime64[ms]	2023-01-01 00:00:00, 2023-01-02 00:00:00, 2023-01-03 00:00:00
date_key	int32	20230101, 20230102, 20230103
year	int32	2023, 2024, 2025
quarter	int32	1, 2, 3
month_num	int32	1, 2, 3
month_name	str	Jan, Feb, Mar
year_month	str	2023-01, 2023-02, 2023-03
month_sort	int32	202301, 202302, 202303
abs_month	int32	24277, 24278, 24279
day_of_month	int32	1, 2, 3
is_month_end	bool	False, True
iso_week	int64	52, 1, 2
window_month_idx	int32	-8, -7, -6
in_analysis_window	bool	False, True
is_holiday_br	bool	False, True
is_holiday_mx	bool	False, True
is_holiday_us	bool	False, True
season_tag	str	year_end_peak, carnaval, regular
is_partial_period	bool	False, True

dim_device

one row per device family · 7 rows

column	dtype	sample values
device_key	int64	1, 2, 3
device_family	str	mobile_ios, mobile_android, web
device_class	str	mobile, web, ctv
is_ctv	bool	False, True

dim_market

one row per market · 3 rows

column	dtype	sample values
market_key	int64	1, 2, 3
market_id	str	BR, MX, US
currency_code	str	BRL, MXN, USD
market_active_from_date_key	int64	20230901, 20240601, 20241201
timezone	str	America/Sao_Paulo, America/Mexico_City, America/New_York
is_comparable_base	bool	True, False

dim_payment_method

one row per payment method · 5 rows

column	dtype	sample values
method_key	int64	1, 2, 3
method	str	card, pix, boleto
is_prepaid	bool	False, True
settlement_lag_days_typical	int64	2, 0, 3

dim_plan

one row per tier x billing period x market · 18 rows

column	dtype	sample values
plan_key	int64	1, 2, 3
plan_id	str	Basico-monthly-BR, Basico-monthly-MX, Basico-monthly-US
tier	str	Basico, Standard, Premium
billing_period	str	monthly, annual
market_id	str	BR, MX, US
market_key	int64	1, 2, 3
has_ads	bool	True, False
max_streams	int64	1, 2, 4
max_resolution	str	1080p, 4K
has_downloads	bool	False, True

dim_price_history

one row per tier x market x price-effective period · 11 rows

column	dtype	sample values
price_key	int64	1, 2, 3
tier	str	Basico, Standard, Premium
market_id	str	BR, MX, US
effective_from_date_key	int64	20230901, 20260301
effective_to_date_key	int64	20271231, 20260300
list_price_local	float64	18.9, 89.0, 6.99

dim_subscriber

one row per subscriber (account with >=1 paid start) · 115,728 rows

column	dtype	sample values
subscriber_key	int64	1, 2, 3
subscriber_id	str	S1000000, S1000001, S1000002
market	str	BR, MX, US
market_key	int64	1, 2, 3
cohort_month	str	2025-12, 2024-03, 2026-02
first_paid_month_idx	int64	27, 6, 29
signup_date_key	int64	20251220, 20240321, 20260223
acquisition_channel	str	partner_bundle, paid_social, direct
channel_key	int64	6, 4, 1
first_tier	str	Standard, Premium, Basico
current_tier	str	Standard, Premium, Basico
billing_period	str	monthly, annual
primary_device_family	str	smart_tv, mobile_ios, streaming_stick
crm_lifecycle_stage	str	new, power_user, engaged
is_incentivised	bool	False, True
is_comparable_base	bool	True, False
is_l4l_cohort	bool	False, True

dim_support_reason

one row per unified support reason category · **6 rows**

column	dtype	sample values
support_reason_key	int64	1, 2, 3
reason_category	str	billing_payment, app_playback, app_crash
reason_group	str	billing, app_tech, account

fact_ad_revenue

one row per market x day · **2,557 rows**

column	dtype	sample values
date_key	int64	20230901, 20230902, 20230903
month_idx	int64	0, 1, 2
market	str	BR, MX, US
market_key	int64	1, 2, 3
impressions_served	int64	1123, 1388, 1319
fill_rate	float64	0.867, 0.765, 0.865
ecpm_usd	float64	11.31, 8.58, 8.95
ad_revenue_usd	float64	12.71, 11.91, 11.82

fact_app_performance_daily

one row per market x date x device family x app version · **15,342 rows**

column	dtype	sample values
market	str	BR, MX, US
local_date	datetime64[us]	2023-09-01 00:00:00, 2023-09-02 00:00:00, 2023-09-03 00:00:00
device_family	str	console, mobile_android, mobile_ios
device_key	int64	6, 2, 1
app_version	str	2.0.4, 2.1.4, 2.2.4
month_idx	int64	0, 1, 2
sessions	int64	48, 265, 168
crashes	int64	0, 1, 2
screen_views	int64	316, 1739, 1095
crash_free_rate	float64	1.0, 0.9962264150943396, 0.9940476190476191
date_key	int64	20230901, 20230902, 20230903

fact_billing_attempt

one row per billing attempt · **1,141,115 rows**

column	dtype	sample values
txn_id	str	pg_204913177311, pg_243834280277, pg_200064501858
subscriber_key	int64	24, 246, 272
market	str	BR, MX, US
month_idx	int32	0, 1, 2
date_key	int64	20230930, 20231031, 20231130
attempt_ts_utc	datetime64[us, UTC]	2023-09-21 00:00:00+00:00, 2023-09-11 00:00:00+00:00, 2023-09-17 00:00:00+00:00
method	str	carrier_billing, card, pix
method_key	int64	5, 1, 2
status	str	approved, failed, refunded
type	str	charge, refund, retry
attempt_number	int64	1, 2, 3
is_dunning	bool	False, True
failure_reason	str	, not_paid, expired
amount_local	float64	26.46, 18.9, 39.04
amount_usd	float64	5.3411, 3.8151, 7.8805
processing_cost_local	float64	2.3814, 1.16734, 0.9481
processing_cost_usd	float64	0.4807, 0.2356, 0.1914

fact_content_cost

one row per title x month · **123,876 rows**

column	dtype	sample values
content_key	int64	1, 2, 3
month_idx	int32	0, 1, 2
amort_usd	float64	0.0, 319.37, 83.72
cash_spend_usd	float64	0.0, 11178.02, 1925.01
date_key	int64	20230930, 20231031, 20231130

fact_engagement_device_month

one row per record · **504 rows**

column	dtype	sample values
market	str	BR, MX, US
device_family	str	console, mobile_android, mobile_ios
device_key	int64	6, 2, 1
month_idx	int64	0, 1, 2
streamed_hours	float64	1538.8311666666666, 3819.2058055555553, 6926.048250000001
play_starts	int64	2229, 5600, 10100
video_start_failures	int64	36, 60, 109
playback_errors	int64	12, 25, 63
rebuffer_ratio	float64	0.011813559322033897, 0.011583167495854063, 0.011925649350649352
cdn_gb	float64	5527.191487304687, 13725.237560546875, 24991.49345410156
viewing_subs	int64	197, 492, 807
market_key	int64	1, 2, 3
vsf_rate	float64	0.01615, 0.01071, 0.01079
playback_error_rate	float64	0.00538, 0.00446, 0.00624
date_key	int64	20230930, 20231031, 20231130

fact_engagement_month

one row per subscriber x month with streaming activity · 1,146,571 rows

column	dtype	sample values
subscriber_key	int64	-1, 24, 246
account_ref	str	\$1031759, \$1037109, \$1068598
month_idx	int64	0, 1, 2
streamed_hours	float64	74.15, 30.589999999999996, 104.0
active_days	int64	19, 9, 18
distinct_titles	int64	56, 33, 69
playback_errors	int64	1, 0, 2
video_start_failures	int64	0, 2, 3
ctv_days	float64	12.0, 13.0, 14.0
pct_days_ctv	float64	0.632, 1.333, 0.667
below_healthy_engagement	bool	False, True
date_key	int64	20230930, 20231031, 20231130

fact_finance_month

one row per month x market x P&L line · 780 rows

column	dtype	sample values
month_idx	int64	0, 1, 2
market	str	BR, MX, US
pnl_line	str	subscription_revenue, advertising_revenue, partner_revenue
amount_usd	float64	2722.6, 353.73, 122.52
market_key	int64	1, 2, 3
date_key	int64	20230930, 20231031, 20231130

fact_fx_rate

one row per month x currency · 108 rows

column	dtype	sample values
month_idx	int64	0, 1, 2
currency	str	BRL, MXN, USD
avg_rate	float64	4.954, 4.8952, 4.9136
eop_rate	float64	4.9761, 4.9183, 4.9177
date_key	int64	20230930, 20231031, 20231130

fact_marketing_spend

one row per week x market x channel x campaign · 2,016 rows

column	dtype	sample values
week_date_key	int64	20230901, 20230908, 20230915
month_idx	int64	0, 1, 2
market	str	BR, MX, US
market_key	int64	1, 2, 3
channel	str	affiliate, direct, organic
channel_key	int64	5, 1, 2
campaign_id	str	cmp_BR_affiliate_2023Q3, cmp_BR_direct_2023Q3, cmp_BR_organic_2023Q3
spend_usd	float64	264.01, 245.15, 226.3
spend_local	float64	264.01, 245.15, 226.3
impressions	int64	66003, 61288, 56574
clicks	float64	543.0, 504.0, 465.0
signups_attributed	int64	16, 25, 23

fact_subscription_event

one row per subscription lifecycle event (movement ledger) · **265,110 rows**

column	dtype	sample values
event_id	str	ent_874752370545, ent_663015193664, ent_667510900029
subscriber_key	int64	1, 2, 3
subject	str	S1000000, S1000001, S1000002
event_type	str	paid_start, churn, cancel_request
event_ts_utc	datetime64[us, UTC]	2025-12-27 00:00:00+00:00, 2026-04-06 00:00:00+00:00, 2026-04-23 00:00:00+00:00
event_date_key	int64	20251227, 20260406, 20260423
month_idx	int32	27, 31, 6
from_tier	str	Premium, Standard, Basico
to_tier	str	Standard, Premium, Basico
reason_code	str	voluntary, too_expensive, temporary_break

fact_subscription_month

one row per subscriber x month while subscribed (folded from the ledger) · **1,216,330 rows**

column	dtype	sample values
subscriber_key	int64	1, 2, 3
subscriber_id	str	S1000000, S1000001, S1000002
month_idx	int64	27, 28, 29
status	str	active, churned
tier	str	Standard, Premium, Basico
billing_period	str	monthly, annual
market	str	BR, MX, US
mrr_local	float64	26.46, 29.9, 0.0
tenure_months	int64	0, 1, 2
is_new	bool	True, False
is_reactivation	bool	False, True
is_voluntary_churn	bool	False, True
is_involuntary_churn	bool	False, True
mrr_usd	float64	4.9495, 4.9367, 5.5673
date_key	int64	20251231, 20260131, 20260228

fact_support_ticket

one row per support ticket · 79,215 rows

column	dtype	sample values
ticket_id	str	v1-0-741029, v1-0-70719851, v1-0-50785554
subscriber_key	int64	31794, 8220, 7874
market	str	BR, MX, US
month_idx	int32	0, 1, 2
created_date_key	int64	20230918, 20230916, 20230927
contact_channel	str	chat, phone, email
reason_category	str	account_access, app_playback, billing_payment
reason_group	str	account, app_tech, billing
support_reason_key	int64	4, 2, 1
csat_score	float64	1.0, 0.72, 0.38
resolution_hours	float64	12.76, 10.72, 0.8
source_version	str	v1, v2

fact_targets_month

one row per market x month x KPI · 432 rows

column	dtype	sample values
market	str	BR, MX, US
market_key	int64	1, 2, 3
month_idx	int64	0, 1, 2
kpi	str	paid_active_subscribers, mrr_usd, gross_monthly_churn_pct
target_value	float64	545.0, 2949.29, 4.0
date_key	int64	20230930, 20231031, 20231130

fact_viewing_daily

one row per subscriber x local date x device family (partitioned by month) · 13,485,904 rows (across 36 monthly parts)

column	dtype	sample values
subscriber_key	int64	-1, 24, 246
account_ref	str	S1031759, S1037109, S1068598
local_date	datetime64[us]	2023-09-01 00:00:00, 2023-09-02 00:00:00, 2023-09-05 00:00:00
device_family	str	mobile_android, mobile_ios, web
device_key	int64	2, 1, 3
month_idx	int64	0
streamed_minutes	float64	207.51833333333335, 160.55666666666667, 487.94666666666666
play_starts	int64	5, 11, 8
distinct_titles	int64	3, 2, 8
video_start_failures	int64	0, 1, 2
playback_errors	int64	0, 1, 2
rebuffer_ratio	float64	0.0084, 0.0107, 0.0069
cdn_gb	float64	4.791995117187501, 3.860876953125, 9.749
date_key	int64	20230901, 20230902, 20230905

Voxa+ — ETL Pipeline

Document status: Phase 4 deliverable · describes `etl/` as built. **Companion:** `docs/02_data_architecture.md` (the spec), `docs/03_data_dictionary.md` (auto-generated column reference), `docs/08_data_quality.md` (issue handling & limitations).

```
python -m pip install -r etl/requirements.txt
python etl/run_pipeline.py           # raw → staging → curated → dq_checks    (~28 min at full scale)
python etl/run_pipeline.py --from 02 # staging onward
python etl/gen_data_dictionary.py    # regenerate docs/03 from data/curated/
```

Everything is driven by one `SEED` in `etl/config.py`. `POP_SCALE` (env `VOXA_POP_SCALE`) scales the simulated population — `1.0` produces ~116k sign-ups / ~66k active subscribers at window end; smaller values are used for fast calibration (churn/retention rates are scale-invariant).

Runtime at `POP_SCALE=1.0` on a Windows/OneDrive workstation: raw ≈ 19 min (≈ 13.5 M subscriber-day viewing rows across 36 monthly Parquet files), staging ≈ 2.5 min, curated ≈ 7 min, DQ checks ≈ 4 s.

1. Design rules that shaped the code

Rule	How it is enforced
Deterministic	<code>utils.rng("stage", "topic", ...)</code> derives an independent NumPy <code>Generator</code> from <code>SEED</code> + a stable hash of the name parts. Every random draw names its own stream, so adding a step never shifts another step's numbers.
Idempotent	Each script wipes and rebuilds only its own layer. Re-running with the same config yields byte-identical output.
Layered	Staging reads only <code>data/raw/</code> ; curated reads only <code>data/staging/</code> ; the semantic model reads only <code>data/curated/</code> . Never upward.
Crash-safe on OneDrive	<code>utils._atomic_write</code> writes to a temp file in the same folder then <code>os.replace</code> , retrying ~5× with backoff on <code>PermissionError</code> / <code>OSError</code> (WinError 5/32).
Every fix is counted	Staging and curated push named counters into <code>data/quality/dq_fragments/*.json</code> ; <code>dq_checks.py</code> assembles them into <code>dq_report.md</code> .

2. Stage 0 — the generative world model (`etl/gen/`)

`01_generate_raw.py` first builds a **clean, internally-consistent picture of the business**, then dirties it. The clean model is three modules:

2.1 `gen/world.py` — subscribers, month by month

Every per-subscriber attribute is a full-length NumPy buffer indexed by subscriber row. The simulation loops over the 36 window-months and, each month, vectorises over the currently active set:

- Acquisition** (`_acquisition_plan`) — Poisson monthly sign-ups per market with a launch ramp for MX/US, organic growth, and calendar seasonality; each new subscriber draws a channel (from the pre- or post- `CHANNEL_SHIFT_IDX` mix), tier, billing period, a persistent primary device and connected-TV viewing share, an engagement propensity, a price sensitivity, an incentivised-sign-up flag, and a payment method.
- Engagement** — monthly streamed hours from a tier base × propensity × tenure decay × seasonality × content-gap factor, with a log-normal shock and an "inactive this month" tail; `below_healthy` = < 5 h.
- Billing & dunning** — monthly (or annual-anchor) charge at the plan-price schedule; authorisation by method; failures enter a 3-retry dunning sequence; exhausted dunning becomes **involuntary churn**.
- Voluntary churn hazard** — a tenure base hazard × multipliers for tier, market, annual lock-in, engagement (low-engagement × 1.5, otherwise a mild slope), and incentivised early tenure; then **additive terms from** `config.py` \$9 (see 2.4). A uniform draw against the clipped hazard decides the churn.
- Plan changes / pause / resume / reactivation** — small monthly probabilities; each emits a lifecycle event.
- Recording** — one `subscription_month` row per active, paused, **or churned-this-month** subscriber; MRR recognised at the monthly price (annual = list ÷ 12).

Outputs: `subscribers`, `subscription_month`, `subscription_events` (the movement ledger), `billing_attempts`, `viewing_month`.

2.2 `gen/content.py`

~3,500 titles with language-appropriate names (PT/ES/EN), genre, an originals flag, a release vintage (some pre-window), licence windows per market (a few BR licences expire mid-window), a tentpole release schedule, and a per-title amortisation + lumpy cash-spend schedule calibrated so content amortisation lands near ~55 % of subscription revenue.

2.3 `gen/economics.py`

FX as a clamped random walk (BRL ≈ 4.9–5.5, MXN ≈ 17–20 per USD); marketing spend derived from sign-up volume × CAC-by-channel with impressions/clicks; SSP ad revenue from Básico viewing × fill × eCPM; support tickets driven by billing failures + app issues + a reason mix; a monthly GL roll-up (`fact_finance_month` source) that ties to the operational aggregates within tolerance.

2.4 The churn-shift mechanism

`config.py` §9 holds a small set of **tunable effect sizes** that make company-wide gross churn step up at window-month 34 (from ~4.1 % to ~6.8 %, a 1.66× "near-doubling") as an **emergent property** of several overlapping parameter changes rather than one switch. The factors are intentionally *not* enumerated in `docs/01` — the point of the Phase 10 investigation is to recover them from the curated data. `docs/walkthrough.md` records what they are for maintainers.

3. Stage 1 — raw (`01_generate_raw.py` → `data/raw/<source>/`)

Serialises the clean world into 13 source-shaped exports, injecting every data-quality issue Q01–Q15 and counting each into `dq_fragments/raw_injection.json`.

Source folder	Files	Deliberate mess
ketch/	<code>plan_catalog.xlsx</code> , <code>invoices__YYYY-MM.csv</code> , <code>subscriptions_snapshot__YYYY-MM.csv</code>	Excel-serial dates + text prices; overlapping monthly invoice ranges; a daily full-dump snapshot (3 copies/month) that drifts from the ledger ; plan/market free-text.
pagstream/	<code>transactions__YYYY-MM.csv</code> , <code>settlement__YYYY-MM.csv</code>	semi-structured meta JSON; amount gross-for-some / net-of-fee-for-others; negative refunds; ~0.3 % null account_id .
voxaid/	<code>accounts.csv</code> (UTF-8 BOM), <code>device_registrations.csv</code>	free-text market; GDPR-deleted accounts omitted though still referenced; unnormalised device names.
entitlement/	<code>events.jsonl</code>	at-least-once delivery → ~1.8 % duplicate event_id ; shuffled order.
bonsai/	<code>crm_contacts.csv</code> (Latin-1), <code>winback_membership.csv</code>	mojibake names; free-text lifecycle stage; birth_year 0/1900; early cohorts absent.
trackpad/	<code>app_events__YYYY-MM.parquet</code>	session-rollup per account/day/device; duplicate event_id ; device/version noise; UTC.
playlog/	<code>playlog__YYYY-MM.parquet</code>	watch time seconds → milliseconds from window-month 20; CDN bytes MB vs GB by edge provider; a batch of negative watch_time ; asset_id s not in the catalogue.
reelbase/	<code>titles.xlsx</code> , <code>licence_windows.csv</code>	encoding damage; mixed DD/MM/YYYY and Excel-serial dates; missing genre; is_original as Y/N/1/0/".
contentfin/	<code>amort_schedule.xlsx</code> , <code>cash_milestones.xlsx</code>	parentheses negatives; month columns as Excel serials; implied currency.
adbridge/	<code>spend_weekly.csv</code> , <code>attribution.csv</code>	channel-name noise; MX rows carry MXN in the spend_usd column; clicks holds impressions on partner rows; attribution has no account key.
voxaads/	<code>ssp_daily_revenue.csv</code>	MM/DD/YYYY dates; duplicate day rows.
helpline/ + helpdesk/	<code>v1_tickets.csv</code> , <code>v2_tickets.csv</code>	schema break at the window-month-13 migration: different column names, reason taxonomy, id format, CSAT scale (1–5 vs 0–10); some tickets duplicated across the cutover.
ledger/	<code>gl_trial_balance__YYYY-MM.csv</code> , <code>fx_rates.csv</code>	free-text account names; a few FX months missing.

`data/raw/_truth/` also holds a handful of clean frames (`subscription_month`, `gl`, ...) that `dq_checks.py` reconciles against — never read by staging or curated.

4. Stage 2 — staging (`02_clean_stage.py` → `data/staging/stg_*`)

One typed `stg_*` table per raw entity. **No cross-source joins, no metric definitions.**

Fix class	Technique
Date parsing	Vectorised <code>parse_date_series</code> — explicit <code>pd.to_datetime(format=...)</code> per known style (<code>iso</code> , <code>dmy</code> , <code>mdy</code> , <code>%d-%b-%y</code>), Excel-serial branch via <code>EXCEL_EPOCH</code> + <code>to_timedelta</code> .
Money parsing	Vectorised <code>parse_money_series</code> — strip non-numeric, decide decimal separator per row by which of <code>,</code> / <code>.</code> appears last, parentheses → negative. Invoices parse per <code>currency</code> group (BRL = 1.234,56; USD/MXN = 1,234.56).
Encoding repair	<code>ftfy.fix_text</code> for mojibake; read the Latin-1 CRM file with the right codec; <code>utf-8-sig</code> to drop the BOM.
De-duplication	<code>drop_duplicates</code> on <code>event_id</code> / <code>invoice_id</code> / <code>beacon_id</code> ; the daily full-dump snapshot is collapsed to the latest row per account per file .
Unit normalisation	PlayLog watch time → seconds by era + magnitude; CDN bytes MB→GB by <code>edge_provider</code> ; CSAT → 0–1 (±5 for v1, ±10 for v2).
Category harmonisation	<code>norm_series</code> maps free text to a controlled vocabulary via <code>etl/vocab/*.csv</code> (<code>.str.strip()</code> . <code>map(dict)</code>); unmatched → explicit <code>"unknown"</code> .
Mislabelled columns	AdBridge MX <code>spend_usd</code> → <code>spend_local</code> (MXN) then recompute USD; partner-row <code>clicks</code> →null. PagStream <code>amount</code> kept as <code>amount_reported</code> + <code>amount_is_trusted</code> = <code>False</code> — curated re-derives the billed amount from the plan-price schedule.
Schema unification	Helpline v1 + Helpdesk v2 mapped to one <code>stg_support_tickets</code> schema and one reason taxonomy via <code>vocab/support_reason_crosswalk.csv</code> , tagged <code>source_version</code> ; cutover duplicates dropped.
Timezones	All event timestamps → tz-aware UTC.

Performance note. The first implementation used `Series.map(python_fn)` for every parse and ran ~45 min at full scale on the 2–4 M-row Ketch tables and the 13.5 M-row PlayLog/ Trackpad loop. Rewriting the hot paths as the vectorised `*_series` helpers above (and replacing per-row `json.loads` on the PagStream `meta` blob with vectorised regex extract) brought staging to ~2.5 min.

The big two (`stg_playlog` , `stg_trackpad_app_events`) are written **partitioned by month** under `data/staging/<name>/YYYY-MM.parquet` so neither staging nor curated has to hold the full 13.5 M rows in memory.

5. Stage 3 — curated (`03_build_curated.py` → `data/curated/`)

Conformed Kimball star: **10 dimensions + 13 facts** (+ `fact_viewing_daily` partitioned by month). Written as Parquet **and** a CSV mirror.

Key build logic:

- **dim_date first**, spanning **2023-01-01** → **2027-12-31** — deliberately wider than the fact window so no stray "(blank)" date member appears when annual renewals or settlement land past the last window month. Carries `window_month_idx` , `in_analysis_window` , per-market holiday flags, `season_tag` , `is_partial_period` .
- **Surrogate keys** minted 1..n per dimension; business keys kept as `*_id` . Sentinels: `-1` unknown, `0` unresolved. Facts referencing a missing dimension get a sentinel and a counted DQ row — rows are **never dropped** for referential problems.
- **dim_subscriber** joins VoxalD accounts + first-paid month from the ledger + first/ current tier + a billing-period detection (annual if any invoice $\approx \geq 6 \times$ the subscriber's median invoice) + CRM lifecycle. Acquisition channel is **sampled from each market's monthly channel mix** (the AdBridge attribution feed has no account key — see limitations). `is_comparable_base` (= Brazil) and `is_l4l_cohort` (first-paid $\leq$ PY end) stamped here.
- **fact_subscription_month is folded from the movement ledger**. Events are sorted `subject` , `month_idx` , `event-type-priority` , `timestamp` (priority so `paid_start` precedes a same-month `churn` despite random within-month timestamps), then replayed per subject into active spells `[start, end, is_reactivation, tier, churn_kind]` . Monthly rows are emitted per spell with `tenure_months` , `is_new` / `is_reactivation` on the first month and `is_voluntary_churn` / `is_involuntary_churn` on the churn month. The Ketch snapshot is used **only** to quantify drift, never as input.
- **fact_billing_attempt** takes the billed amount from the plan-price schedule (working around the PagStream `amount` mislabel), keeps status/method/attempt/dunning from the gateway, and adds `processing_cost_*` from the method fee model.
- **fact_viewing_daily** aggregates the PlayLog beacons to subscriber $\times$ local date $\times$ device family per month; `fact_engagement_month` rolls that up to subscriber $\times$ month with `below_healthy_engagement` , `active_days` , `pct_days_ctv` .
- **USD everywhere**. Every `_local` money column gets a `_usd` sibling at the month's average FX rate from `fact_fx_rate` (= 1 for USD).
- **fact_targets_month** builds the plan (`PY actual  $\times$  (1 + planned growth)` per market, flat target churn) local-then-converted.

6. DQ gate (`dq_checks.py`)

Runs last; **fails the pipeline (non-zero exit) on any hard check**. Assembles `data/quality/dq_report.md` from all fragments + the results.

Hard checks (structural / accounting): no null surrogate keys in facts; every event resolves to a real or sentinel subscriber; the subscriber base **roll-forward identity** (`start` + `new` + `reactivation` – `churn` = `end` , ≤ 2 % monthly — actual worst 0.00 %); cohort retention ≤ 100 % and non-increasing; `fact_viewing_daily` device keys $\subseteq$ `dim_device` ; `dim_date` spans beyond the fact window.

Soft checks (plausibility / calibration, warn only): unresolved-subscriber share on billing ≤ 1.2 %; MRR positive & rising; pre-shift blended gross churn in 3.6–4.6 % (actual 4.11 %); peak (idx 33–35) in 6.4–8.6 % (actual 6.82 %); peak/pre ratio 1.6–2.1 $\times$ (actual 1.66 $\times$); pre-shift streamed hours 25–70 h (47.6 h); below-healthy share 8–28 % (11.1 %); GL subscription revenue vs curated MRR ≤ 15 % mean rel. (4.0 %).

Latest run: **6 hard PASS, 0 soft warnings.**

Voxa+ — Data Model

Document status: Phase 4 deliverable · the curated star as built in `data/curated/` . **Companion:** `docs/03_data_dictionary.md` (every column + samples), `docs/06_dax_measures.md` (Phase 7). The Power BI semantic model (Phase 6) is generated from these schemas.

Grain of the mart: **the subscriber-month** for retention and revenue, with event-, attempt-, and day-grain facts beneath it for the diagnostic pages.

1. Bus matrix (facts × conformed dimensions)

Fact ↓ / Dimension →	date	market	subscriber	plan	channel	device	payment method	content	support reason	app version
fact_subscription_month	•	•	•	O(tier)	O(via sub)	O(via sub)				
fact_subscription_event	•	O(via sub)	•	O(from/to tier)						
fact_billing_attempt	•	•	•	O(tier)			•			
fact_viewing_daily	•	O(via sub)	•			•		O(content_key)		
fact_engagement_month	•	O(via sub)	•			O(pct_days_ctv)				
fact_app_performance_daily	•	•				•				•
fact_support_ticket	•	•	•						•	
fact_marketing_spend	• (week)	•			•					
fact_ad_revenue	•	•								
fact_content_cost	• (month)	O(alloc)						•		
fact_fx_rate	• (month)	O(currency)								
fact_finance_month	• (month)	•		O(pnl_line)						
fact_targets_month	• (month)	•								

• direct relationship · O derived / attribute-level / bridged in DAX.

2. Facts — grain, measures, degenerate columns

Fact	Grain	Rows (full run)	Additive measures	Non-additive / flags	Degenerate dims
fact_subscription_month	subscriber × window-month while subscribed	1,216,330	mrr_local, mrr_usd	status, tenure_months, is_new, is_reactivation, is_voluntary_churn, is_involuntary_churn	tier, billing_period, market
fact_subscription_event	one lifecycle event	265,110	count	event_type, from_tier, to_tier, reason_code	event_id, subject
fact_billing_attempt	one billing attempt	1,141,115	amount_local/usd, processing_cost_local/usd	status, type, attempt_number, is_dunning, failure_reason	txn_id, method
fact_viewing_daily	subscriber × local date × device family	13,485,904	streamed_minutes, play_starts, distinct_titles, video_start_failures, playback_errors, cdn_gb	rebuffer_ratio (avg)	—
fact_engagement_month	subscriber × window-month with streaming	1,146,571	streamed_hours, active_days, distinct_titles, playback_errors, video_start_failures, ctv_days	pct_days_ctv, below_healthy_engagement	—
fact_app_performance_daily	market × date × device family × app version	15,342	sessions, crashes, screen_views	crash_free_rate	app_version
fact_support_ticket	one ticket	79,215	count, resolution_hours	csat_score, is flags, source_version	ticket_id, contact_channel
fact_marketing_spend	ISO week × market × channel × campaign	2,016	spend_usd, spend_local, impressions, clicks, signups_attributed	—	campaign_id
fact_ad_revenue	market × day	2,557	impressions_served, ad_revenue_usd	fill_rate, ecpm_usd	—
fact_content_cost	title × window-month	123,876	amort_usd, cash_spend_usd	—	—
fact_fx_rate	window-month × currency	108	—	avg_rate, eop_rate	currency
fact_finance_month	window-month × market × P&L line	780	amount_usd	—	pnl_line
fact_targets_month	market × window-month × KPI	432	target_value	—	kpi

month_idx (0-based window month) and date_key (YYYYMMDD int, month-end for monthly-grain facts) appear on every fact for the relationship to dim_date.

3. Dimensions — key attributes

Dimension	Key	Rows	Attributes used on the report
dim_date	date_key	1,826	year_month, month_sort (sort), window_month_idx, in_analysis_window, season_tag, is_holiday_{br,mx,us}, is_partial_period
dim_market	market_key	3	market_id (BR/MX/US), currency_code, market_active_from_date_key, is_comparable_base
dim_subscriber	subscriber_key	115,728	market, cohort_month, first_paid_month_idx, acquisition_channel, first_tier, current_tier, billing_period, primary_device_family, crm_lifecycle_stage, is_comparable_base, is_l4l_cohort
dim_plan	plan_key	18	tier, billing_period, market_id, has_ads, max_streams, max_resolution, has_downloads
dim_price_history	price_key	11	tier, market_id, effective_from/to_date_key, list_price_local — range table, used in DAX, not a relationship
dim_channel	channel_key	7	channel, channel_group, is_low_quality
dim_device	device_key	7	device_family, device_class (mobile/web/ctv), is_ctv
dim_payment_method	method_key	5	method, is_prepaid, settlement_lag_days_typical
dim_content	content_key	3,500	title, content_type, primary_genre, language, is_original, release_date_key, runtime_min
dim_support_reason	support_reason_key	6	reason_category, reason_group (billing / app_tech / content / account / other)
dim_app_version	app_version_key	15	app_version, major, is_v3

dim_reporting_currency (a disconnected BR L/MXN/USD slicer table) and the P&L-line and funnel helper tables are added in **Phase 6** with the semantic model, not in the curated files.

4. Relationships

All single-direction (dim 1 → * fact), single cross-filter, unless noted.

From (dim)	To (fact)	Key	Notes
dim_date[date_key]	every fact [date_key]	date_key	monthly facts join on month-end key
dim_market[market_key]	fact_subscription_month, fact_billing_attempt, fact_support_ticket, fact_app_performance_daily, fact_marketing_spend, fact_ad_revenue, fact_finance_month, fact_targets_month	market_key	facts that only carry the market string get market_key added in Phase 6
dim_subscriber[subscriber_key]	fact_subscription_month, fact_subscription_event, fact_billing_attempt, fact_engagement_month, fact_viewing_daily, fact_support_ticket	subscriber_key	the hub of the model; market/channel/tier/device slice through here
dim_device[device_key]	fact_viewing_daily, fact_app_performance_daily	device_key	
dim_content[content_key]	fact_viewing_daily, fact_content_cost	content_key	orphan beacons/rows carry content_key = -1
dim_payment_method[method_key]	fact_billing_attempt	method_key	
dim_channel[channel_key]	fact_marketing_spend	channel_key	subscriber-level channel is an attribute of dim_subscriber
dim_support_reason[support_reason_key]	fact_support_ticket	support_reason_key	
dim_app_version[app_version_key]	fact_app_performance_daily	app_version → key in Phase 6	

Role-playing dates. Only one active dim_date relationship per fact (the event/period date). Secondary dates (release_date_key, market_active_from_date_key, effective_from_date_key) are attributes, resolved in DAX with USERELATIONSHIP only if a visual needs them — kept inactive to avoid ambiguous filter paths.

Deliberately not related. `dim_price_history` (a from/to range — DAX looks up the price in effect for a `tier / market / month`), and `fact_fx_rate` (joined in measures on `month_idx + currency`, or pre-applied as the `_usd` columns).

5. Comparable Base vs Expansion in the model

`is_comparable_base` lives on `dim_market` and `dim_subscriber` (= Brazil). Measures provide three cuts:

- **Total** — no filter.
- **Comparable Base** — `CALCULATE(..., dim_subscriber[is_comparable_base] = TRUE)`.
- **Expansion** — `Total - Comparable Base` for additive measures; computed within the filter for rates.

`is_l4l_cohort` (first-paid month ≤ PY end) supports a 12-month like-for-like cohort cut that sits entirely inside the pre-shift baseline.

6. Known modelling caveats (see also `docs/08_data_quality.md`)

- Subscriber **acquisition channel** is a sampled attribute (the attribution source has no account key) — channel-level retention is directionally correct, not row-exact.
- **Billing amount** comes from the plan-price schedule, not the gateway `amount` column.
- The **last window month** is flagged `is_partial_period` (partner settlement lag) — trend visuals should annotate or exclude it.
- `fact_finance_month` is a modelled roll-up; it ties to operational MRR within ~4 % but is not a general ledger.

Voxa+ — DAX Measure Library

Auto-generated from `etl/pbi_measures.py` on 2026-09-02 by `etl/gen_dax_doc.py`. Do not edit by hand.

113 measures in the `_Measures` table, injected into the semantic model by `powerbi/gen_semantic_model.py`.

Conventions

- Monetary measures are computed in USD from the curated `*_usd` columns, then multiplied by `[FX Rate to Reporting]` so the **Reporting Currency** slicer re-expresses them. `[Currency Symbol]` is available for labels and the narrative.
- Time intelligence uses `dim_date[abs_month]` (a linear `year*12 + month` index), so it needs **no marked date table** and never calls `SAMEPERIODLASTYEAR`. Prior year = `abs_month - 12`.
- Comparable Base** = Brazil (`dim_subscriber[is_comparable_base]`); **Expansion** = Mexico + US.
- Churn, CAC and payback are **lower-is-better**: their `vs Target` measures are good when negative, and the KPI cards invert the delta colour.

00 Currency

FX Rate to Reporting

format `0.0000` · hidden

USD-to-selected-reporting-currency factor for the current date context.

```
VAR cur = SELECTEDVALUE ( 'Reporting Currency'[Currency], "USD" )
RETURN
    IF (
        cur = "USD",
        1,
        CALCULATE (
            AVERAGE ( fact_fx_rate[avg_rate] ),
            REMOVEFILTERS ( 'Reporting Currency' ),
            fact_fx_rate[currency] = cur
        )
    )
```

Currency Symbol

Symbol of the selected reporting currency, for labels and the narrative.

```
SELECTEDVALUE ( 'Reporting Currency'[Symbol], "$" )
```

01 Subscribers

Paid Active Subscribers

format `#,##0`

Mean monthly count of subscribers with an in-force paid subscription. North Star.

```
AVERAGEX (
    VALUES ( dim_date[window_month_idx] ),
    CALCULATE (
        DISTINCTCOUNT ( fact_subscription_month[subscriber_key] ),
        fact_subscription_month[status] = "active"
    )
)
```

Paid Active Subscribers (EoP)

format `#,##0`

Active subscribers in the last month of the analysis window (end of period).

```
VAR lastm =
    CALCULATE ( MAX ( dim_date[window_month_idx] ),
        REMOVEFILTERS ( dim_date ), dim_date[in_analysis_window] = TRUE )
RETURN
    CALCULATE ( [Paid Active Subscribers],
        FILTER ( ALL ( dim_date ), dim_date[window_month_idx] = lastm ) )
```

Active at Start of Month

format #,##0 · hidden

```
CALCULATE (
    DISTINCTCOUNT ( fact_subscription_month[subscriber_key] ),
    fact_subscription_month[status] = "active",
    FILTER ( ALL ( dim_date ), dim_date[window_month_idx] = MAX ( dim_date[window_month_idx] ) - 1 )
)
```

New Subscribers

format #,##0

First paid billing in the period.

```
CALCULATE ( COUNTROWS ( fact_subscription_month ), fact_subscription_month[is_new] = TRUE )
```

Reactivations

format #,##0

```
CALCULATE ( COUNTROWS ( fact_subscription_month ), fact_subscription_month[is_reactivation] = TRUE )
```

Voluntary Churned Subscribers

format #,##0

```
CALCULATE ( COUNTROWS ( fact_subscription_month ), fact_subscription_month[is_voluntary_churn] = TRUE )
```

Involuntary Churned Subscribers

format #,##0

```
CALCULATE ( COUNTROWS ( fact_subscription_month ), fact_subscription_month[is_involuntary_churn] = TRUE )
```

Gross Churned Subscribers

format #,##0

```
[Voluntary Churned Subscribers] + [Involuntary Churned Subscribers]
```

Net Adds

format #,##0

```
[New Subscribers] + [Reactivations] - [Gross Churned Subscribers]
```

02 Churn

Gross Monthly Churn %

format 0.0%

Mean of the monthly (gross churned / start-of-month active) ratios in the period. Watch metric.

```

AVERAGEX (
  FILTER ( VALUES ( dim_date[window_month_idx] ),
    dim_date[window_month_idx] ≥ 0 && dim_date[window_month_idx] ≤ 35 ),
  DIVIDE ( CALCULATE ( [Gross Churned Subscribers] ), CALCULATE ( [Active at Start of Month] ) )
)

```

Voluntary Churn %

format 0.0%

```

AVERAGEX (
  FILTER ( VALUES ( dim_date[window_month_idx] ),
    dim_date[window_month_idx] ≥ 0 && dim_date[window_month_idx] ≤ 35 ),
  DIVIDE ( CALCULATE ( [Voluntary Churned Subscribers] ), CALCULATE ( [Active at Start of Month] ) )
)

```

Involuntary Churn %

format 0.0%

```

AVERAGEX (
  FILTER ( VALUES ( dim_date[window_month_idx] ),
    dim_date[window_month_idx] ≥ 0 && dim_date[window_month_idx] ≤ 35 ),
  DIVIDE ( CALCULATE ( [Involuntary Churned Subscribers] ), CALCULATE ( [Active at Start of Month] ) )
)

```

Pre-Shift Baseline Churn %

format 0.0%

Blended gross monthly churn over window-months 25-33 (the stable pre-shift band).

```

CALCULATE (
  [Gross Monthly Churn %],
  REMOVEFILTERS ( dim_date ),
  dim_date[window_month_idx] ≥ 24 && dim_date[window_month_idx] ≤ 32
)

```

Churn vs Baseline (pp)

format +0.0%;-0.0%;0.0%

Percentage points above (or below) the pre-shift baseline. Higher is worse.

```

[Gross Monthly Churn %] - [Pre-Shift Baseline Churn %]

```

Churn vs Baseline (x)

format 0.00"x"

```

DIVIDE ( [Gross Monthly Churn %], [Pre-Shift Baseline Churn %] )

```

Net MRR Retention %

format 0.0%

MRR from the cohort active 12 months ago, now vs then (incl. up/downgrades and churn).

```

VAR wlast =
    CALCULATE ( MAX ( dim_date[abs_month] ),
        REMOVEFILTERS ( dim_date ), dim_date[in_analysis_window] = TRUE )
VAR c = MIN ( MAX ( dim_date[abs_month] ), wlast )
VAR baseSubs =
    CALCULATETABLE (
        VALUES ( fact_subscription_month[subscriber_key] ),
        FILTER ( ALL ( dim_date ), dim_date[abs_month] = c - 12 ),
        fact_subscription_month[status] = "active"
    )
VAR startMRR =
    CALCULATE ( SUM ( fact_subscription_month[mrr_usd] ),
        FILTER ( ALL ( dim_date ), dim_date[abs_month] = c - 12 ), baseSubs )
VAR endMRR =
    CALCULATE ( SUM ( fact_subscription_month[mrr_usd] ),
        FILTER ( ALL ( dim_date ), dim_date[abs_month] = c ), baseSubs )
RETURN DIVIDE ( endMRR, startMRR )

```

03 Cohorts

Cohort Size

format #,##0

Subscribers at tenure month 0 in the current filter context (use with a cohort slicer).

```
CALCULATE ( DISTINCTCOUNT ( fact_subscription_month[subscriber_key] ), fact_subscription_month[tenure_months] = 0 )
```

Active Subscribers at Tenure

format #,##0 · hidden

```
CALCULATE ( DISTINCTCOUNT ( fact_subscription_month[subscriber_key] ), fact_subscription_month[status] = "active" )
```

Retention %

format 0.0%

Share of a cohort still active — put cohort_month on rows and tenure_months on columns.

```

DIVIDE (
    [Active Subscribers at Tenure],
    CALCULATE ( [Cohort Size], REMOVEFILTERS ( fact_subscription_month[tenure_months] ) )
)

```

M1 Retention %

format 0.0%

```

VAR c0 = [Cohort Size]
VAR c1 = CALCULATE ( DISTINCTCOUNT ( fact_subscription_month[subscriber_key] ),
    fact_subscription_month[tenure_months] = 1, fact_subscription_month[status] = "active",
    REMOVEFILTERS ( fact_subscription_month[tenure_months] ) )
RETURN DIVIDE ( c1, c0 )

```

M3 Retention %

format 0.0%

```

VAR c0 = [Cohort Size]
VAR c3 = CALCULATE ( DISTINCTCOUNT ( fact_subscription_month[subscriber_key] ),
    fact_subscription_month[tenure_months] = 3, fact_subscription_month[status] = "active",
    REMOVEFILTERS ( fact_subscription_month[tenure_months] ) )
RETURN DIVIDE ( c3, c0 )

```

M6 Retention %

format 0.0%

```
VAR c0 = [Cohort Size]
VAR c6 = CALCULATE ( DISTINCTCOUNT ( fact_subscription_month[subscriber_key] ),
    fact_subscription_month[tenure_months] = 6, fact_subscription_month[status] = "active",
    REMOVEFILTERS ( fact_subscription_month[tenure_months] ) )
RETURN DIVIDE ( c6, c0 )
```

M12 Retention %

format 0.0%

```
VAR c0 = [Cohort Size]
VAR c12 = CALCULATE ( DISTINCTCOUNT ( fact_subscription_month[subscriber_key] ),
    fact_subscription_month[tenure_months] = 12, fact_subscription_month[status] = "active",
    REMOVEFILTERS ( fact_subscription_month[tenure_months] ) )
RETURN DIVIDE ( c12, c0 )
```

04 Engagement

Streamed Hours

format #,##0

```
SUM ( fact_engagement_month[streamed_hours] )
```

Streaming Subscribers

format #,##0 · hidden

```
DISTINCTCOUNT ( fact_engagement_month[subscriber_key] )
```

Monthly Active Rate %

format 0.0%

Share of paid subscribers who streamed at least once in the month.

```
DIVIDE ( [Streaming Subscribers], [Paid Active Subscribers] )
```

Viewing Hours per Active Sub

format 0.0

```
DIVIDE ( [Streamed Hours], [Streaming Subscribers] )
```

% Subs Below Healthy Engagement

format 0.0%

Streaming subscribers with < 5 hours this month — a leading indicator of churn.

```
DIVIDE (
    CALCULATE ( DISTINCTCOUNT ( fact_engagement_month[subscriber_key] ),
        fact_engagement_month[below_healthy_engagement] = TRUE ),
    [Streaming Subscribers]
)
```

Titles Started per Active Sub

format 0.0

```
DIVIDE ( SUM ( fact_engagement_month[distinct_titles] ), [Streaming Subscribers] )
```

Avg % Days on Connected TV

format 0.0%

```
AVERAGE ( fact_engagement_month[pct_days_ctv] )
```

Streamed Hours (by Device)

format #,##0

```
SUM ( fact_engagement_device_month[streamed_hours] )
```

Video Start Failure Rate

format 0.00%

Play attempts that never start, by device family / market.

```
DIVIDE ( SUM ( fact_engagement_device_month[video_start_failures] ), SUM ( fact_engagement_device_month[play_starts] ) )
```

Playback Error Rate

format 0.00%

```
DIVIDE ( SUM ( fact_engagement_device_month[playback_errors] ), SUM ( fact_engagement_device_month[play_starts] ) )
```

Rebuffer Ratio

format 0.00%

```
AVERAGE ( fact_engagement_device_month[rebuffer_ratio] )
```

05 App Quality

App Sessions

format #,##0

```
SUM ( fact_app_performance_daily[sessions] )
```

Crashes

format #,##0

```
SUM ( fact_app_performance_daily[crashes] )
```

Crash-Free Session Rate %

format 0.00%

```
DIVIDE ( [App Sessions] - [Crashes], [App Sessions] )
```

% Sessions on App v3

format 0.0%

```
DIVIDE (
  CALCULATE ( [App Sessions], dim_app_version[is_v3] = TRUE ),
  [App Sessions]
)
```

06 Payments

Billing Attempts

format #,##0

```
COUNTROWS ( fact_billing_attempt )
```

Approved Attempts

format #,##0

```
CALCULATE ( [Billing Attempts], fact_billing_attempt[status] = "approved" )
```

Failed Attempts

format #,##0

```
CALCULATE ( [Billing Attempts], fact_billing_attempt[status] = "failed" )
```

Payment Failure Rate %

format 0.0%

First-attempt billing failures (excludes dunning retries).

```
DIVIDE (
    CALCULATE ( [Failed Attempts], fact_billing_attempt[is_dunning] = FALSE ),
    CALCULATE ( [Billing Attempts], fact_billing_attempt[is_dunning] = FALSE )
)
```

Authorization Rate %

format 0.0%

Approved / attempted — slice by payment method.

```
DIVIDE ( [Approved Attempts], [Billing Attempts] )
```

Dunning Recovery Rate %

format 0.0%

```
DIVIDE (
    CALCULATE ( [Approved Attempts], fact_billing_attempt[is_dunning] = TRUE ),
    CALCULATE ( [Billing Attempts], fact_billing_attempt[is_dunning] = TRUE )
)
```

Processing Cost

format #,##0

```
SUM ( fact_billing_attempt[processing_cost_usd] ) * [FX Rate to Reporting]
```

07 Revenue

MRR

format #,##0

```
CALCULATE ( SUM ( fact_subscription_month[mrr_usd] ), fact_subscription_month[status] = "active" ) * [FX Rate to Reporting]
```

ARPU

format #,##0.00

```
DIVIDE ( [MRR], [Paid Active Subscribers] )
```

Subscription Revenue

format #,##0

```
CALCULATE ( SUM ( fact_finance_month[amount_usd] ), fact_finance_month[pnl_line] = "subscription_revenue" ) * [FX Rate to Reporting]
```

Advertising Revenue

format #,##0

```
CALCULATE ( SUM ( fact_finance_month[amount_usd] ), fact_finance_month[pnl_line] = "advertising_revenue" ) * [FX Rate to Reporting]
```

Partner Revenue

format #,##0

```
CALCULATE ( SUM ( fact_finance_month[amount_usd] ), fact_finance_month[pnl_line] = "partner_revenue" ) * [FX Rate to Reporting]
```

Total Revenue

format #,##0

```
[Subscription Revenue] + [Advertising Revenue] + [Partner Revenue]
```

Content Amortisation

format #,##0

```
-1 * CALCULATE ( SUM ( fact_finance_month[amount_usd] ), fact_finance_month[pnl_line] = "content_amortization" ) * [FX Rate to Reporting]
```

CDN Cost

format #,##0

```
-1 * CALCULATE ( SUM ( fact_finance_month[amount_usd] ), fact_finance_month[pnl_line] = "streaming_delivery" ) * [FX Rate to Reporting]
```

Payment Cost

format #,##0

```
-1 * CALCULATE ( SUM ( fact_finance_month[amount_usd] ), fact_finance_month[pnl_line] = "payment_processing" ) * [FX Rate to Reporting]
```

Support Cost

format #,##0

```
-1 * CALCULATE ( SUM ( fact_finance_month[amount_usd] ), fact_finance_month[pnl_line] = "customer_support" ) * [FX Rate to Reporting]
```

Marketing Spend

format #,##0

```
SUM ( fact_marketing_spend[spend_usd] ) * [FX Rate to Reporting]
```

Gross Profit

format #,##0

```
[Total Revenue] - [Content Amortisation] - [CDN Cost] - [Payment Cost] - [Support Cost]
```

Gross Margin %

format 0.0%

```
DIVIDE ( [Gross Profit], [Total Revenue] )
```

Contribution Margin

format #,##0

```
[Gross Profit] - [Marketing Spend]
```

Contribution Margin % (T12M)

format 0.0%

```
VAR c = MAX ( dim_date[abs_month] )
VAR flt = FILTER ( ALL ( dim_date ), dim_date[abs_month] > c - 12 && dim_date[abs_month] ≤ c )
RETURN DIVIDE ( CALCULATE ( [Contribution Margin], flt ), CALCULATE ( [Total Revenue], flt ) )
```

08 Unit Economics

Blended CAC

format #,##0.00

```
DIVIDE ( [Marketing Spend], [New Subscribers] )
```

Expected Lifetime (months)

format 0.0

```
DIVIDE ( 1, [Gross Monthly Churn %] )
```

LTV

format #,##0.00

```
MAX ( 0, [ARPU] * [Gross Margin %] * [Expected Lifetime (months)] )
```

LTV to CAC

format 0.00

```
DIVIDE ( [LTV], [Blended CAC] )
```

CAC Payback (months)

format 0.0

```
DIVIDE ( [Blended CAC], [ARPU] * [Gross Margin %] )
```

09 Time Intelligence

Paid Active Subscribers PY

format #,##0 · hidden

```
CALCULATE ( [Paid Active Subscribers], TREATAS ( SELECTCOLUMNS ( VALUES ( dim_date[abs_month] ), "abs_month", dim_date[abs_month] - 1 ) ) )
```

Paid Active Subscribers YoY %

format +0.0%;-0.0%;0.0%

```
DIVIDE ( [Paid Active Subscribers] - [Paid Active Subscribers PY], ABS ( [Paid Active Subscribers PY] ) )
```

MRR PY

format #,##0 · hidden

```
CALCULATE ( [MRR], TREATAS ( SELECTCOLUMNS ( VALUES ( dim_date[abs_month] ), "abs_month", dim_date[abs_month] - 12 ), dim_date[abs_mo
```

MRR YoY %

format +0.0%;-0.0%;0.0%

```
DIVIDE ( [MRR] - [MRR PY], ABS ( [MRR PY] ) )
```

Total Revenue PY

format #,##0 · hidden

```
CALCULATE ( [Total Revenue], TREATAS ( SELECTCOLUMNS ( VALUES ( dim_date[abs_month] ), "abs_month", dim_date[abs_month] - 12 ), dim_d
```

Total Revenue YoY %

format +0.0%;-0.0%;0.0%

```
DIVIDE ( [Total Revenue] - [Total Revenue PY], ABS ( [Total Revenue PY] ) )
```

Net Adds PY

format #,##0 · hidden

```
CALCULATE ( [Net Adds], TREATAS ( SELECTCOLUMNS ( VALUES ( dim_date[abs_month] ), "abs_month", dim_date[abs_month] - 12 ), dim_date[a
```

Net Adds YoY %

format +0.0%;-0.0%;0.0%

```
DIVIDE ( [Net Adds] - [Net Adds PY], ABS ( [Net Adds PY] ) )
```

Gross Monthly Churn % PY

format 0.0% · hidden

```
CALCULATE ( [Gross Monthly Churn %], TREATAS ( SELECTCOLUMNS ( VALUES ( dim_date[abs_month] ), "abs_month", dim_date[abs_month] - 12
```

Gross Monthly Churn % YoY %

format +0.0%;-0.0%;0.0%

```
DIVIDE ( [Gross Monthly Churn %] - [Gross Monthly Churn % PY], ABS ( [Gross Monthly Churn % PY] ) )
```

MRR MoM %

format +0.0%;-0.0%;0.0%

```
DIVIDE ( [MRR] - CALCULATE ( [MRR], FILTER ( ALL ( dim_date ), dim_date[abs_month] = MAX ( dim_date[abs_month] ) - 1 ) ), CALCULATE (
```

Total Revenue YTD

format #,##0

```
CALCULATE (
    [Total Revenue],
    FILTER ( ALL ( dim_date ),
        dim_date[year] = MAX ( dim_date[year] )
        && dim_date[abs_month] ≤ MAX ( dim_date[abs_month] ) )
)
```

Total Revenue MAT

format #,##0

```
CALCULATE (
    [Total Revenue],
    FILTER ( ALL ( dim_date ),
        dim_date[abs_month] > MAX ( dim_date[abs_month] ) - 12
        && dim_date[abs_month] ≤ MAX ( dim_date[abs_month] ) )
)
```

10 Comparable Base

Paid Active Subscribers (Comparable Base)

format #,##0

```
CALCULATE ( [Paid Active Subscribers], dim_subscriber[is_comparable_base] = TRUE )
```

Paid Active Subscribers (Expansion)

format #,##0

```
CALCULATE ( [Paid Active Subscribers], dim_subscriber[is_comparable_base] = FALSE )
```

Net Adds (Comparable Base)

format #,##0

```
CALCULATE ( [Net Adds], dim_subscriber[is_comparable_base] = TRUE )
```

Net Adds (Expansion)

format #,##0

```
CALCULATE ( [Net Adds], dim_subscriber[is_comparable_base] = FALSE )
```

MRR (Comparable Base)

format #,##0

```
CALCULATE ( [MRR], dim_subscriber[is_comparable_base] = TRUE )
```

MRR (Expansion)

format #,##0

```
CALCULATE ( [MRR], dim_subscriber[is_comparable_base] = FALSE )
```

Gross Monthly Churn % (Comparable Base)

format 0.0%

```
CALCULATE ( [Gross Monthly Churn %], dim_subscriber[is_comparable_base] = TRUE )
```

Gross Monthly Churn % (Expansion)

format 0.0%

```
CALCULATE ( [Gross Monthly Churn %], dim_subscriber[is_comparable_base] = FALSE )
```

11 Targets

Target Paid Active Subscribers

format #,##0

```
CALCULATE ( SUM ( fact_targets_month[target_value] ), fact_targets_month[kpi] = "paid_active_subscribers", FILTER ( ALL ( dim_date ),
```

Target MRR

format #,##0

```
CALCULATE ( SUM ( fact_targets_month[target_value] ), fact_targets_month[kpi] = "mrr_usd", FILTER ( ALL ( dim_date ), dim_date[abs_mo
```

Target Gross Churn %

format 0.0%

```
DIVIDE ( CALCULATE ( AVERAGE ( fact_targets_month[target_value] ), fact_targets_month[kpi] = "gross_monthly_churn_pct" ), 100 )
```

Paid Active Subscribers vs Target %

format +0.0%;-0.0%;0.0%

Higher is better.

```
DIVIDE ( [Paid Active Subscribers (EoP)] - [Target Paid Active Subscribers], [Target Paid Active Subscribers] )
```

MRR vs Target %

format +0.0%;-0.0%;0.0%

Higher is better.

```
DIVIDE ( [MRR (Latest)] - [Target MRR], [Target MRR] )
```

Gross Churn vs Target (pp)

format +0.0%;-0.0%;0.0%

LOWER is better — a negative value is good (below target).

```
[Gross Monthly Churn %] - [Target Gross Churn %]
```

Blended CAC vs Target %

format +0.0%;-0.0%;0.0%

Target CAC US\$22. LOWER is better.

```
DIVIDE ( [Blended CAC] - 22, 22 )
```

Subscribers Lost vs Plan (since shift)

format #,##0

Cumulative gap to plan from window-month 34 onward — the board scoreboard.

```
CALCULATE (
    [Target Paid Active Subscribers] - [Paid Active Subscribers (EoP)],
    REMOVEFILTERS ( dim_date ),
    dim_date[window_month_idx] ≥ 33
)
```

12 Acquisition

% New via Low-Quality Channels

format 0.0%

Affiliate + partner-bundle share of new subscribers.

```
DIVIDE (
  CALCULATE ( [New Subscribers],
    dim_subscriber[acquisition_channel] IN { "affiliate", "partner_bundle" } ),
  [New Subscribers]
)
```

% New Incentivised

format 0.0%

```
DIVIDE ( CALCULATE ( [New Subscribers], dim_subscriber[is_incentivised] = TRUE ), [New Subscribers] )
```

Marketing Impressions

format #,##0

```
SUM ( fact_marketing_spend[impressions] )
```

Signups Attributed

format #,##0

```
SUM ( fact_marketing_spend[signups_attributed] )
```

Cohort M3 Retention (context)

format 0.0%

Alias of M3 Retention % — put acquisition_channel on the axis for channel quality.

```
[M3 Retention %]
```

13 Support

Support Tickets

format #,##0

```
COUNTROWS ( fact_support_ticket )
```

Tickets per 1,000 Subs

format #,##0

```
DIVIDE ( [Support Tickets], [Paid Active Subscribers] ) * 1000
```

CSAT

format 0.00

Mean post-contact satisfaction rescaled to 1-5.

```
AVERAGE ( fact_support_ticket[csat_score] ) * 5
```

% Tickets App/Tech

format 0.0%

```
DIVIDE ( CALCULATE ( [Support Tickets], fact_support_ticket[reason_group] = "app_tech" ), [Support Tickets] )
```

% Tickets Billing

format 0.0%

```
DIVIDE ( CALCULATE ( [Support Tickets], fact_support_ticket[reason_group] = "billing" ), [Support Tickets] )
```

14 Narrative

Exec Insight

Dynamic narrative that updates with the slicers; ships in the file (replaces the Smart Narrative visual).

```
VAR churn = [Gross Monthly Churn % (Latest)]
VAR base = [Pre-Shift Baseline Churn %]
VAR ratio = DIVIDE ( churn, base )
VAR adds = [Net Adds (Latest)]
VAR br = CALCULATE ( [Gross Monthly Churn % (Latest)], dim_subscriber[market] = "BR" )
VAR mx = CALCULATE ( [Gross Monthly Churn % (Latest)], dim_subscriber[market] = "MX" )
VAR us = CALCULATE ( [Gross Monthly Churn % (Latest)], dim_subscriber[market] = "US" )
VAR worst =
    SWITCH ( TRUE (),
        br ≥ mx && br ≥ us, "Brazil",
        mx ≥ br && mx ≥ us, "Mexico",
        "the United States" )
RETURN
    "Gross monthly churn is " & FORMAT ( churn, "0.0%" ) & ". "
    & IF (
        ratio ≥ 1.35,
        "That is about " & FORMAT ( ratio, "0.0" ) & "x the pre-shift baseline of "
        & FORMAT ( base, "0.0%" ) & " and the highest of the three markets is "
        & worst & " (" & FORMAT ( MAX ( MAX ( br, mx ), us ), "0.0%" ) & "). ",
        "That is close to the pre-shift baseline of " & FORMAT ( base, "0.0%" ) & ". " )
    & "Net adds in the latest month: " & FORMAT ( adds, "###0" ) & ". "
    & "Reporting currency: " & [Currency Symbol] & "."
```

15 Latest Month

Gross Monthly Churn % (Latest)

format 0.0%

Value for the last month in the current filter context — for summary KPI cards.

```
VAR lastm =
    CALCULATE ( MAX ( dim_date[window_month_idx] ),
        REMOVEFILTERS ( dim_date ), dim_date[in_analysis_window] = TRUE )
RETURN CALCULATE ( [Gross Monthly Churn %],
    FILTER ( ALL ( dim_date ), dim_date[window_month_idx] = lastm ) )
```

Churn vs Baseline (x) (Latest)

format 0.00"x"

Value for the last month in the current filter context — for summary KPI cards.

```
VAR lastm =
    CALCULATE ( MAX ( dim_date[window_month_idx] ),
        REMOVEFILTERS ( dim_date ), dim_date[in_analysis_window] = TRUE )
RETURN CALCULATE ( [Churn vs Baseline (x)],
    FILTER ( ALL ( dim_date ), dim_date[window_month_idx] = lastm ) )
```

Net Adds (Latest)

format ###0

Value for the last month in the current filter context — for summary KPI cards.

```
VAR lastm =
    CALCULATE ( MAX ( dim_date[window_month_idx] ),
        REMOVEFILTERS ( dim_date ), dim_date[in_analysis_window] = TRUE )
RETURN CALCULATE ( [Net Adds],
    FILTER ( ALL ( dim_date ), dim_date[window_month_idx] = lastm ) )
```

MRR (Latest)

format #,##0

Value for the last month in the current filter context — for summary KPI cards.

```
VAR lastm =  
    CALCULATE ( MAX ( dim_date[window_month_idx] ),  
        REMOVEFILTERS ( dim_date ), dim_date[in_analysis_window] = TRUE )  
RETURN CALCULATE ( [MRR],  
    FILTER ( ALL ( dim_date ), dim_date[window_month_idx] = lastm ) )
```

LTV to CAC (Latest)

format 0.00

Value for the last month in the current filter context — for summary KPI cards.

```
VAR lastm =  
    CALCULATE ( MAX ( dim_date[window_month_idx] ),  
        REMOVEFILTERS ( dim_date ), dim_date[in_analysis_window] = TRUE )  
RETURN CALCULATE ( [LTV to CAC],  
    FILTER ( ALL ( dim_date ), dim_date[window_month_idx] = lastm ) )
```

Net MRR Retention % (Latest)

format 0.0%

Value for the last month in the current filter context — for summary KPI cards.

```
VAR lastm =  
    CALCULATE ( MAX ( dim_date[window_month_idx] ),  
        REMOVEFILTERS ( dim_date ), dim_date[in_analysis_window] = TRUE )  
RETURN CALCULATE ( [Net MRR Retention %],  
    FILTER ( ALL ( dim_date ), dim_date[window_month_idx] = lastm ) )
```


Voxa+ — Visual Identity

Document status: Phase 5 deliverable. **Assets:** `assets/brand.py` (palette + type, the single source of truth), `assets/gen_logo.py` (emits the mark, wordmark and page backgrounds), `powerbi/theme/theme.json` (passes `pbir theme validate`), `powerbi/assets/` (generated PNGs + `_page_heights.json`).

1. Tone

Corporate, restrained, board-facing. Voxa+ is a streaming service, so the identity is media-adjacent — a play glyph, a confident violet — but this is a **churn diagnosis for the board**, not a marketing page. No gradients on data, no glow, no more than one accent colour competing for attention at a time.

2. Colour

Role	Hex	Use
Brand / primary series	#4B2E83 violet	logo, current-value bars/lines, <code>tableAccent</code>
Header band / darkest ink	#241348	the page-background band; chart drop-shadow tint
Brand wash	#EDE9F5	selected slicer item, hover
Accent (one only)	#E6A23C amber	"look here" — current period, the hairline under the band, KPI emphasis
Comparison	#5B7FB0 steel blue	prior year, target, secondary series
Categorical ramp	violet · steel · amber · #3E9C93 · #8E5EA8 · #6B7280 · #C77CA6 · #4C6EA6	channels, tiers, markets, genres
Page ground	#F2F1F7	page + outspace background
Card / surface	FFFFFF	
Neutral fill	#E5E3EF	"rest" bars, disabled
Gridline / hairline	#E7E5EF / #D9D6E4	1 px, quiet
Good / Neutral / Bad	#2E7D5B / #8A8A99 / #C0453B	KPI deltas, waterfall sentiment, conditional formatting
Diverging (CF)	min #C0453B · mid #EFC48A · max #2E7D5B	retention %, vs-target %

Churn is lower-is-better. KPI delta and vs-target colours are inverted for churn, CAC and payback (a *decrease* is #2E7D5B good); the waterfall bridges use the same #2E7D5B / #C0453B sentiment pair as the KPI deltas.

3. Type

Class	Face	Size	Colour
Page title	Segoe UI Semibold	30	#1E1A33
Visual title	Segoe UI Semibold	12	#1E1A33
KPI callout value	Segoe UI Semibold	30	#1E1A33
Body / data label	Segoe UI	9	#3A3646
Axis / muted	Segoe UI	9	#77738A
On the dark band	Segoe UI / Semibold	15–20	#F4F2FA, muted #C6BEDD

4. The signature element (every page)

A full-page **background PNG** (`powerbi/assets/bg_<page>.png`, generated by `gen_logo.py`), because a bound image visual does not render reliably through a hand-generated PBIR:

1. **Deep-violet band** (#241348, 60 px) with the **Voxa+ wordmark** left and a quiet "Subscriber Churn Diagnosis" label right.
2. **Amber hairline** (#E6A23C, 3 px) directly under the band.
3. On the light body: the **page title + one-line business question**, then a faint rule.

On top of the PNG, every page places, in this order: - **Currency** slicer + **Market** slicer (top-right, on the band zone). - **Analysis-Period** date slicer (joined on `order / month` keys, never an Advanced `dim_date` filter — guardrail). - One row of **six KPI cards**: a single-measure value card + a small "vs Target"

delta card, both transparent / no border / no shadow, sitting on a **white rounded frame** `shape` (corner radius 20, no shadow). Delta green if better / red if worse, inverted for lower-is-better metrics. - Two content rows on the **3-30-300** gradient (glance → scan → study).

Per-page heights (`powerbi/assets/_page_heights.json`): **Executive = 900**, all others **720** — so adding visuals to the exec page never stretches the rest.

5. Theme (`powerbi/theme/theme.json`)

`visualStyles` for ~20 visual types + `page`. Highlights:

- Charts: white surface, 8 px rounded border, **soft drop shadow** (violet-tinted, 88 % transparent) — **only on charts**, never on cards, frames, slicers, shapes or images.
- Gridlines: horizontal only, `#E7E5EF`, 1 px; category axis gridlines off; axis titles off by default.
- Data labels **on** at theme level for bar / column / line / combo / waterfall; **off** for stacked and area (dense series get a per-visual override in Phase 8).
- `card`: category label shown small and muted, value 30 pt semibold, no background/border (the frame `shape` provides the surface).
- `slicer`: no header, no border, transparent; title muted 9 pt; single-select default.
- `waterfallChart`: sentiment colours pinned to good/bad/violet.
- `tableEx` / `pivotTable`: horizontal rules only, violet-tinted header fill, 9 pt.

6. Regenerating

```
python assets/gen_logo.py          # mark, wordmark, 7 page backgrounds, _page_heights.json
pbir theme validate powerbi/theme/theme.json
```


Voxa+ — Data Quality

Document status: Phase 4 deliverable. **Generated companion:** `data/quality/dq_report.md` (rebuilt every pipeline run by `etl/dq_checks.py`), fed by `data/quality/dq_fragments/*.json`.

Every issue in the Phase-2 catalogue (Q01–Q15) is **injected** by `01_generate_raw.py`, **handled** in staging or curated, **counted** into a DQ fragment, and — where it is an invariant — **asserted** by `dq_checks.py`. Counts below are from the shipped full-scale run (`SEED=20260902`, `POP_SCALE=1.0`).

1. Issue → handling → evidence

#	Issue	Injected (count)	Handled in	Handling	Asserted
Q01	Multiple date formats + Excel serials	47 serial cells + all date columns	02 parse_date_series	per-style to_datetime ; serial = EXCEL_EPOCH + days ; 125,638 serials converted, 140,024 text dates parsed	dates_unparseable = 0
Q02	Money as locale strings, parentheses negatives	1,173,363 cells	02 parse_money_series	decimal separator decided per row; (x) → -x (10,190); 1,161,624 parsed	no money column left as object dtype
Q03	Encoding damage (Latin-1 mojibake, BOM)	114,784 names + 3,500 titles + 1 BOM	02 ftfy.fix_text , utf-8-sig , Latin-1 read	117,061 strings repaired, BOM stripped	no Ã/Â runs in dim_content.title
Q04	Duplicate rows (events, invoices, daily full-dump)	4,771 + 303,064 + 303,064 events; 32,248 invoice; 2,442,218 snapshot copies	02 drop_duplicates ; per-file snapshot collapse	623,966 dup events dropped; 32,310 invoice lines; 2,442,218 snapshot rows collapsed	fact_subscription_event unique on event_id ; invoices unique on invoice_id
Q05	Master rows missing (orphan subscriber / content)	465 GDPR-deleted + 101,533 orphan content	03 sentinel keys -1 / 0 + count	1,025 event rows, 8,003 billing rows, 101,533 viewing rows → sentinel	every fact FK resolves (real or sentinel); no null surrogate keys
Q06	Mislabelled columns (clicks =impressions; spend_usd =MXN; amount gross/net)	336 + 648 + 478,434	02 column correction; 03 re-derive amount	partner clicks →null, MX spend restored to MXN + USD recomputed (984 corrected); billing amount taken from the plan-price schedule	marketing clicks ≤ impressions
Q07	Snapshot-vs-ledger drift	14,652 drift rows	03 snapshot folded from the ledger	Ketch snapshot used only to measure drift; curated status = ledger status by construction	base roll-forward identity holds (worst 0.00 % monthly)
Q08	Sign errors (negative refunds, negative watch, negative recovery)	11,546 + 33,569	02 sign normalisation	refunds stored with consistent sign; 33,569 negative watch rows → abs + flagged	fact_viewing_daily.streamed_minutes ≥ 0
Q09	Unit inconsistency (watch s→ms, CDN MB/GB, CSAT /5 /10)	12,954,757 + 6,736,491 + 79,380	02 unit normalisation	ms era + magnitude → seconds; edge-b bytes ÷ 1024; CSAT rescaled to 0-1	watch minutes/sub-day ≤ 1440; csat_score ∈ [0,1]
Q10	Categorical noise (channel/plan/market/device free text)	~1.77 M values touched	02 norm_series + etl/vocab/*.csv	1,766,151 values normalised; 0 fell through to unknown this run	all dim_* category columns ∈ controlled vocab
Q11	Schema break (Helpline v1 → Helpdesk v2)	5,344 v1 + 74,036 v2; 164 cutover dupes	02 schema unification + crosswalk	one stg_support_tickets schema + one taxonomy; 165 cutover dupes dropped	fact_support_ticket single schema; taxonomy ∈ dim_support_reason
Q12	Late / partial / restating files	4 FX months missing; last month partial	02 FX forward-fill; dim_date.is_partial_period	4 FX months filled; window-month 36 flagged partial	last-month completeness flagged, not silently averaged
Q13	Null keys (guest billing rows, retired plan)	3,424 null account_id	02 flag; 03 sentinel	3,424 flagged; billing rows with unresolved subscriber = 0.70 %	soft check ≤ 1.2 %
Q14	Outliers (test campaign, 30h+/day, annual as one line)	test-campaign flag; annual invoices	03 quarantine / explode / caps	annual plans explode to 12 monthly recognitions; test campaign excluded from fact_marketing_spend	marketing KPIs exclude the quarantined campaign
Q15	Timezone / month-boundary bleed	UTC events vs local billing	02 local-day derivation per market	local_date from tz-converted timestamp; month_idx from the local day	monthly viewing totals reconcile to daily within tolerance

2. Hard checks (fail the build)

From the latest run — **6/6 PASS**:

Check	Result
No null subscriber_key in fact_subscription_month	PASS
fact_subscription_event subscriber_key resolves or sentinel	PASS
Subscriber base roll-forward identity start + new + reactivation - churn = end ($\leq 2\%$ /month)	PASS — worst 0.00 %
Cohort retention $\leq 100\%$ and non-increasing	PASS — M1 89 % / M6 74 % / M12 64 %
fact_viewing_daily.device_key $\subseteq$ dim_device	PASS
dim_date spans beyond the fact window	PASS

3. Soft checks (plausibility & calibration — warn only)

From the latest run — **0 warnings**:

Check	Value	Band
Billing rows with unresolved subscriber	0.70 %	$\leq 1.2\%$
Pre-shift blended gross monthly churn (idx 24–32)	4.11 %	3.6–4.6 %
Peak blended gross churn (idx 33–35)	6.82 %	6.4–8.6 %
Peak / pre ratio	1.66×	1.6–2.1×
Avg streamed hours / sub-month (pre-shift)	47.6 h	25–70 h
Share below healthy engagement (pre-shift)	11.1 %	8–28 %
GL subscription revenue vs curated MRR (mean rel. error)	4.0 %	$\leq 15\%$

The full churn-spike shape and cohort-retention curve are tabulated in `data/quality/dq_report.md`.

4. Accepted limitations

These are known and left in on purpose; the report and the Phase-10 analysis annotate them.

- Acquisition channel is a sampled attribute.** The AdBridge attribution feed carries no account key, so each subscriber's `acquisition_channel` is drawn from their market's monthly channel mix. Channel-level retention and quality trends are directionally sound but not row-exact; conclusions about acquisition quality are stated as cohort-level, not individual-level.
- Billing amount is taken from the plan-price schedule**, not the gateway `amount` column (Q06 mislabel). Revenue and payment-cost figures are therefore schedule-accurate; the gateway's reported amount is kept as `amount_reported` for audit only.
- The last window month is partial.** `dim_date.is_partial_period` flags window-month 36; partner settlement and some carrier-billing rows lag by a month. Trend visuals should exclude or annotate it.
- Snapshot-vs-ledger drift is measured, not corrected.** `fact_subscription_month` is folded from the movement ledger; the 14,652 drifting Ketch snapshot rows are reported as a DQ metric, not reconciled.
- fact_finance_month is a modelled roll-up**, not a general ledger. It ties to operational MRR within $\sim 4\%$ and is adequate for margin trend and the CFO page, not for statutory reporting.
- M12 cohort retention ($\sim 64\%$) sits a little above the docs/01 benchmark band (38–55 %).** The simulated base carries a $\sim 15\text{--}19\%$ annual-plan mix (locked in) and counts win-backs as re-activations, both of which lift 12-month logo retention. The Phase-10 analysis uses first-spell-only retention where a stricter figure is needed.
- Gross margin is thin and ramps.** Content amortisation is $\sim 90\%+$ of revenue early in the window on a small base and settles near $\sim 55\%$ by window-end (26 % gross margin at window-month 34) — a deliberate scaling-company trajectory, slightly below the docs/01 steady-state band.
- No external / competitor data.** The "Market Context" page compares Voxa+'s own three markets, not the competitive market.

Voxa+ — Report Structure

Document status: Phase 8 deliverable. **Build:** `cd powerbi && bash build_report.sh` (regenerates assets → semantic model → layout manifest → drives pbir to build VoxaChurnDiagnosis.Report). **Layout source of truth:** `powerbi/gen_report_visuals.py` → `powerbi/_report_layout.json`.

The report is **7 pages, one business question each** — the seven stakeholder personas from docs/01 §12. Every page carries the same signature furniture, then two content rows on the 3-30-300 gradient (glance → scan → study).

Signature element (every page)

- 1. **Page-background PNG** (`powerbi/assets/bg_<page>.png`): deep-violet band + **Voxa+** wordmark + a quiet "Subscriber Churn Diagnosis" label, an amber hairline, then the page **title + the business question** on the light body.
- 2. **Slicer row** ($y \approx 150$): **Analysis period** (`dim_date[year_month]`) left; **Market** (`dim_subscriber[market]`) and **Currency** (`Reporting Currency[Currency]`) right.
- 3. **KPI band:** a white rounded frame `shape` (radius 20, no shadow) behind **six value cards**; KPIs with a plan target also carry a small **delta card** beneath, conditionally coloured green / red — **inverted for churn, CAC and payback** (lower is better).
- 4. Report-level filter `dim_date[in_analysis_window] = true` keeps every visual inside the 36-month window.

Executive page height 900; the other six 720 (`powerbi/assets/_page_heights.json`).

Page 1 — Executive Summary

How bad is the churn spike, and is it still going?

KPIs: Paid Subscribers (EoP) · Gross Monthly Churn % · Churn vs Baseline (x) · Net Adds · MRR · LTV : CAC — deltas vs plan target / YoY.

Visual	Type	Fields
Gross monthly churn vs the pre-shift baseline	line	Gross Monthly Churn %, Pre-Shift Baseline Churn % by year_month
Read this	card	Exec Insight (dynamic narrative)
Gross monthly churn by market	clustered column	Gross Monthly Churn % by year_month × market
Voluntary vs involuntary churned subscribers	clustered column	Voluntary / Involuntary Churned Subscribers by year_month
Net subscriber adds by month	column	Net Adds by year_month

Page 2 — Subscriber Retention & Cohorts

Which cohorts, tenures, plans and sources are churning?

KPIs: Gross / Voluntary / Involuntary Churn % · M1 / M6 Retention % · Net MRR Retention %.

Visual	Type	Fields
Cohort retention triangle	matrix	Retention % — rows cohort_month, cols tenure_months
Gross monthly churn by plan tier	bar	Gross Monthly Churn % by current_tier
Gross monthly churn by tenure (months)	line	Gross Monthly Churn % by tenure_months
Churned subscribers by acquisition channel	column	Gross Churned Subscribers by acquisition_channel

Page 3 — Content & Engagement

Did viewing drop before people cancelled?

KPIs: Monthly Active Rate % · Hours / Active Sub · % Below Healthy Engagement · Titles / Active Sub · % Days on CTV · Streamed Hours.

Visual	Type	Fields
Viewing hours per active subscriber	line	Viewing Hours per Active Sub by year_month
Video-start-failure rate by device class	clustered column	Video Start Failure Rate by year_month × device_class
Share of subscribers below healthy engagement	line	% Subs Below Healthy Engagement by year_month
Streamed hours by device family	column	Streamed Hours (by Device) by device_family

Page 4 — Pricing & Plans

Did a price or plan change trigger it?

KPIs: ARPU · Gross Margin % · Gross Monthly Churn % · % Below Healthy Engagement · MRR · Net MRR Retention %.

Visual	Type	Fields
Gross monthly churn by tier over time	clustered column	Gross Monthly Churn % by year_month × current_tier
Plan list-price history	table	dim_price_history tier / market / list price
ARPU by market	clustered column	ARPU by year_month × market
Churned subscribers by tier and month (filter market = BR)	clustered column	Gross Churned Subscribers by year_month × current_tier

Page 5 — Acquisition Quality & Channels

Did the quality of who we acquire shift?

KPIs: New Subscribers · % via Low-Q Channels · % Incentivised · M1 / M3 Retention % · Blended CAC.

Visual	Type	Fields
New subscribers by acquisition channel	stacked column	New Subscribers by year_month × acquisition_channel
M3 retention by acquisition channel	bar	M3 Retention % by acquisition_channel
Share of new subscribers via low-quality channels	line	% New via Low-Quality Channels by year_month
Low-quality-channel share by market	clustered column	% New via Low-Quality Channels by year_month × market

Page 6 — Customer Experience, Billing & App Quality

Payment failures or app problems forcing people out?

KPIs: Payment Failure Rate % · Dunning Recovery % · Involuntary Churn % · Crash-Free Rate % · Tickets / 1k Subs · CSAT.

Visual	Type	Fields
Crash-free session rate by device class	clustered column	Crash-Free Session Rate % by year_month × device_class
Payment failure rate and dunning recovery	line	Payment Failure Rate %, Dunning Recovery Rate % by year_month
Authorization rate by payment method	bar	Authorization Rate % by method
Support contacts per 1,000 subscribers by reason	clustered column	Tickets per 1,000 Subs by year_month × reason_group

Page 7 — Market Context

Is this global, or concentrated in one market?

KPIs: Gross Churn (Total / Comparable Base / Expansion) · Net Adds · MRR · Subscribers Lost vs Plan (since shift).

Visual	Type	Fields
Gross monthly churn by market	clustered column	Gross Monthly Churn % by year_month × market
Net adds by market	clustered column	Net Adds by year_month × market
Comparable Base vs Expansion churn	clustered column	Gross Monthly Churn % (Comparable Base / Expansion) by year_month
Read this	card	Exec Insight

First open in Power BI Desktop (one time)

The `pbir` build validates structure and every field against the on-disk TMDL model, but it cannot render. On first open:

1. **Transform data** → `pDataFolder` → point at `.../voxaplus-churn-diagnosis/data/curated` → **Close & Apply** (imports ~4 M fact rows; a minute or two).
2. **View** → **Themes** → **Browse for themes** → `powerbi/theme/theme.json`.
3. **Model view** → `dim_date` → **Mark as date table** (column `date`).
4. Walk the pages; iterate layout / formatting with `pbir` + `pbir desktop refresh` + `pbir desktop screenshot`. Known open items from `pbir validate --all`: KPI value/delta cards are intentionally compact and layered on the frame (`VISUAL_OVERLAP` / `VISUAL_UNDERSIZED` warnings are expected); a few pages run ~20 visuals (`TOO_MANY_VISUALS`).

Voxa+ — Subscriber Churn: Root-Cause Diagnosis

Analyst deep-dive · prepared for the leadership team · all figures dual-computed in DuckDB SQL and pandas with a parity check on every result (`../parity/parity_report.md` — 14/14 pass).

0. The question and the answer

Question. Gross monthly subscriber churn nearly doubled starting at window-month 34 (Jun 2026) and has stayed elevated. What caused it?

Answer. Three factors that overlapped in time. None explains the spike on its own.

	Factor	Role	Markets
1	An app "v3" connected-TV playback regression (window-month 32)	acute trigger, biggest contributor	all; weakest in the US
2	The Brazil Standard/Premium price increase (window-month 32), landing on an already-disengaged base	amplifier	Brazil only
3	A months-old Mexico cohort-quality drag (newer Mexican subscribers retain ~8 pp worse)	pre-existing drain	Mexico only

Payments and involuntary churn were tested and are **not** involved (§6).

1. The spike — size, persistence, shape (*findings F01, F02*)

Gross monthly churn stepped from a **4.1 %** pre-shift baseline (window-months 25–33) to **6.8 %** across the three elevated months — a **1.66×** increase, and **90 % of it is voluntary**. It has not receded: the three months read 6.85 %, 6.58 %, 7.02 %.

Splitting by market shows the company-wide number is a composite of three different shapes:

Market	Pre-shift	Peak	Behaviour
Brazil	3.4 %	7.0 %	flat, then a clean +3.0 pp step at window-month 34 — acute
Mexico	5.4 %	8.1 %	drifting up from ~window-month 29 (+0.6 pp), then a further step — a problem that got worse
United States	4.1 %	4.4 %	essentially unchanged — the control

A US that barely moves already rules out any explanation that would hit every market equally.

2. Who is leaving, and why now (*findings F03, F04*)

The lift is broad, not narrow. Peak-vs-baseline churn is up by a strikingly even ~+2.7 pp across almost every tenure band, plan tier and acquisition channel. Two exceptions carry information:

- **Annual plans lifted only +1.9 pp** (vs +2.7 pp monthly) — locked-in subscribers were partly shielded.
- **Long-tenure subscribers (25+ months) were hit as hard as brand-new ones** (+3.2 pp) — a cause confined to weak new cohorts could not do that.

A single narrow cause does not produce an even cross-segment lift. This is the first quantitative sign of **multiple overlapping factors**.

The extra churn is engagement-led. For subscribers who cancelled during the elevated months, streamed hours were already sliding beforehand — ~37 h two months out, ~36 h the month before — whereas pre-shift churners left from a flat ~43 h. Value erodes first, the cancellation follows. That points at what suppressed viewing just before the spike (§3), and it explains why a price change bites harder now (§4): people were already getting less.

3. Factor 1 — the connected-TV app regression (*finding F05*)

The Voxa+ "v3" app shipped device-family by device-family, reaching connected-TV last. Connected-TV streaming quality is flat through window-month 31, then:

window-month	31	32	33–36
CTV video-start-failure rate	1.1 %	3.0 %	~3.4 %
CTV playback-error rate	0.6 %	1.4 %	~1.6 %

Mobile and web are unchanged throughout. The connected-TV rates **triple at window-month 32 — one month before the churn step**.

The churn consequence is specific to the affected subscribers:

	Pre-shift	Peak
Not CTV-heavy	4.1 %	5.6 %
CTV-heavy (primary device = smart-TV / stick / console)	4.1 %	8.4 %

Before the shift the two groups churned **identically**. After the CTV rollout, CTV-heavy churn **doubled**. The one-month lag between the quality drop and the churn step is a value → decision delay of roughly one billing cycle. This is the **acute trigger and the largest single contributor**, and — because it touches every market — it is why the company-wide number looks global.

4. Factor 2 — the Brazil price increase, and why it bit (*finding F06*)

Brazil raised Standard and Premium list prices at window-month 32, migrating existing members on their next renewal. Split Brazil Std/Prem subscribers by whether they were engaged or disengaged (below the healthy-engagement line) in the month:

Brazil Std/Prem	Churn before hike	Churn after hike	Lift
Engaged	~0 %	~0 %	≈ 0 pp
Disengaged	23.9 %	42.2 %	+18.3 pp

The price increase had **no measurable effect on engaged subscribers** and a very large effect on **already-disengaged** ones. Because Factor 1 pushed a wave of Brazil subscribers into the disengaged bucket in window-months 32–33, the migration then landed on a larger disengaged pool than it otherwise would have. Price alone did not move the base; **price × prior disengagement produced the Brazil step**. Brazil-only.

5. Factor 3 — Mexico's pre-existing cohort-quality drag (*finding F07*)

Month-3 retention by market and cohort era:

Market	Pre-shift cohorts	Post-shift cohorts (acquired m24–30)
Brazil	~82.8 %	~80 % (–3 pp)
Mexico	~81.8 %	~73 % (–8–9 pp)
United States	~82.8 %	~81 % (–2 pp)

Every market's newer cohorts retain a little worse; **Mexico's are ~8–9 pp worse at month 3** — three times the drop elsewhere. It is **not channel-specific** (within post-shift Mexico, low-quality-channel cohorts retain about the same as the rest), so this is not "the channel mix kept degrading" — it is **Mexican conditions for newly-acquired subscribers deteriorating across the board** from ~window-month 24. This drag was already lifting Mexican churn ~0.6 pp before the visible spike and is still underneath it.

6. What it is *not* — payments (*finding F08*)

Market	First-attempt payment-failure rate (pre → peak)	Dunning recovery (pre → peak)	Involuntary churn (pre → peak)
Brazil	7.6 % → 7.7 %	51 % → 52 %	0.43 % → 0.41 %
Mexico	18.8 % → 18.2 %	37 % → 37 %	1.25 % → 1.30 %
United States	8.0 % → 7.8 %	58 % → 57 %	0.31 % → 0.32 %

Nothing moved. Combined with the 90 % voluntary share (\$1), payments are **ruled out** as a driver of the spike. Mexico's payment economics are structurally poor but that is a standing cost, not a new one.

7. Impact, and the decomposition (*findings F09, F10*)

Impact. The churn step cuts the implied average subscriber lifetime from ~24 months to ~15 — a ~40 % reduction in lifetime value. In the three elevated months, ~5,400 subscribers left over and above the pre-shift rate; at ~\$7.5 ARPU that is on the order of **\$0.5–0.6 M of lifetime recurring revenue already forgone**, compounding monthly. A 40 % lower LTV also cuts the allowable CAC by ~40 %, so the growth plan no longer pencils.

Decomposition. Tag every peak-month voluntary churn with the three factor flags:

Market	Peak voluntary churn	F1 (CTV)	F2 (price×diseng)	F3 (low-q young)	none
Brazil	6,629	54 %	63 %	12 %	15 %
Mexico	4,017	53 %	0 %	23 %	36 %
United States	1,597	52 %	0 %	20 %	39 %

In Brazil, **85 % of the elevated churn carries Factor 1 or Factor 2**. In Mexico, ~64 % carries Factor 1 or Factor 3. The US "39 % none" is just its unchanged ~4 % baseline — the control. The flags overlap, so these are *coverage*, not an independent split; the causal weight sits in the **rate** comparisons (\$3–\$5), which are clean.

8. Recommendations (priority order)

1. **Fix the connected-TV player (Sev-1)**. Roll back / hotfix v3 on smart-TV and streaming-stick; verify video-start-failure returns to ~1 %. Save-campaign CTV-heavy subscribers whose viewing fell after window-month 32.
 2. **Pause the Brazil price migration for low-engagement Standard/Premium subscribers**. Offer a retention path (hold price / downgrade / pause); re-test elasticity after the CTV fix.
 3. **Audit Mexico acquisition and onboarding since mid-2025**. Cap spend on the weakest sources until month-3 retention recovers toward ~82 %.
 4. **Instrument for next time**. Connected-TV QoE on the executive dashboard, a leading "engagement health" trend, and price-change elasticity by engagement band.
-

How this was produced

`python data_analysis/run_analysis.py` runs all ten questions twice — DuckDB SQL over the curated Parquet and pandas over the same files — and asserts the two agree within tolerance before any number is used. SQL is saved under `data_analysis/sql/`, agreed results under `data_analysis/results/`, and the parity log at `data_analysis/parity/parity_report.md`. Per-problem method, evidence and limitations are in `data_analysis/findings/F01 – F10`.

Voxa+ — Build Walkthrough

A running log. Each entry: what was done · why · where to look. Newest at the bottom.

Phase 0 — Scope & scaffold

- Locked scope with the stakeholder (see `README.md`): SVOD service **Voxa+**, 36-month window (Sep 2023 – Aug 2026), churn shift at window-month 34, markets BR/MX/US with MX and US launching mid-window, USD reporting currency + slicer, balanced depth, `.pbip` / PBIR from code, single `SEED`.
- Three stakeholder choices resolved via a decision prompt: **balanced data scale** (~120k cumulative sign-ups, ~70k active at window end, viewing at subscriber-day grain), **market launch months** MX = window-month 10 / US = window-month 16, and **North Star = Paid Active Subscribers** with watch metric Gross Monthly Churn %.
- Created the folder tree (`docs/` `etl/` `data/{raw,staging,curated,quality}/` `powerbi/` `assets/` `portfolio/` `data_analysis/`).

Phase 1 — Company story & KPI framework

- `docs/01_business_context_kpis.md`: company profile, the 36-month window rationale (CY = months 25–36, PY = 13–24), expansion timeline, plan line-up and margin structure, financial definitions (the churn/subscriber spine), North Star, Total vs Comparable Base (= Brazil) vs Expansion, multi-currency handling, seasonality, the seven stakeholder personas → the seven report pages, and the full KPI framework with benchmark ranges → targets.
- Section 8 "Events in the window"** lists 12 neutral, dated business events (cancellation redesign, support migration, ad-tier renegotiation, licence expiries, marketing-mix shift, app "v3" rollout, Brazil price increase, plan-lineup tweak, ...). None is flagged as a cause — they are the candidate context the Phase 10 investigation weighs.
- The root cause is deliberately **absent** from this document (assumption A15).

Phase 2 — Data architecture

- `docs/02_data_architecture.md`: medallion principles (raw → staging → curated; seeded, idempotent, layered), tech stack + run instructions, naming conventions, an inventory of **13 simulated source systems**, the raw-file catalogue with volumes, the staging fix classes, the curated Kimball star (**14 dimensions + 14 facts**), a **15-item data-quality issue catalogue (Q01–Q15)** each mapped injected-in → handled-in → hard-check, and a Mermaid architecture diagram.
- Key design choices carried into build: `fact_subscription_month` is **folded from the movement ledger** (the Ketch snapshot is used only to measure drift); annual plans explode into 12 monthly recognitions; subscriber-level QoE is preserved so engagement can be tied to individual churn; `is_comparable_base` propagated from `dim_subscriber`.

Phase 3 — ETL pipeline

3.0 Structure

- `etl/config.py` — one `SEED`, window dates, launch months, every tunable, plus **section 9 "ROOT-CAUSE EFFECT SIZES"** (see below). `POP_SCALE` (env `VOXA_POP_SCALE`) scales the population for fast calibration; `1.0` → ~117k sign-ups / ~67k active at end.
- `etl/utils.py` — deterministic per-stream RNG (`rng("name", ...)`), atomic + retry-on-lock IO (OneDrive WinError 5/32), messy-format emitters (Excel serials, locale money strings, mojibake, BOM, category jitter), DQ-fragment writer.
- `etl/gen/world.py` — the clean generative world model (subscriber-level monthly simulation: acquisition → engagement → billing/dunning → churn hazard → plan changes / pause / reactivation → the movement ledger). Full-length NumPy buffers per attribute; loop over 36 months, vectorised over the active set.
- `etl/gen/content.py` — catalogue, licence windows, release schedule, amortisation + cash.
- `etl/gen/economics.py` — FX random walk, marketing spend (from acquisition × CAC by channel), SSP ad revenue, support tickets, a monthly GL roll-up.
- `etl/01_generate_raw.py` — builds the world, then serialises it into 13 source-shaped messy exports; every Q01–Q15 injection is counted into `dq_fragments/raw_injection.json`.
- `etl/02_clean_stage.py` — one `stg_*` per raw entity; parsing / encoding repair / de-dup / unit normalisation / category harmonisation; counts every fix.
- `etl/03_build_curated.py` — the conformed star; `fact_subscription_month` replays the ledger per subject (events sorted `subject, month_idx, event-priority, ts` so logical order survives the random within-month timestamps).
- `etl/dq_checks.py` — hard structural + accounting-identity checks (fails the build) plus soft plausibility/calibration checks; assembles `data/quality/dq_report.md`.
- `etl/run_pipeline.py` (`--from`, `--only`), `etl/gen_data_dictionary.py`.

3.1 The root cause (decided here; kept out of `docs/01`)

The window-month-34 gross-churn spike (~4.1% pre-shift → ~6.8% peak, **1.66×** — "nearly doubled") is the product of **three overlapping factors**, encoded as tunable effect sizes in `config.py` §9:

	Factor	Role	Where it shows
F1	App "v3" connected-TV playback regression — the re-platformed player shipped device-by-device, CTV last (smart-TV/stick at window-month 33, console 34). Video-start-failure / rebuffer / crash rise ~2–3× on CTV; CTV-heavy subscribers lose ~30% of viewing hours and carry an extra ~3.5 pp monthly voluntary hazard, lagged one month . US barely affected (APP_V3_CTV_US_SENSITIVITY).	Acute trigger	Content & Engagement (by device), CX/App Quality
F2	Brazil price increase (Std/Prem +13/15% at window-month 32) — migrated on each subscriber's next renewal, spread over months 32–36. Engaged BR subs absorb it (+3 pp); disengaged BR Std/Prem leave (+13.5 pp) — and F1 is what makes many of them disengaged. Basico reacts to the month-33 plan-lineup change.	Amplifier (BR only)	Pricing & Plans (engagement × tier × timing)
F3	Mexico acquisition-quality decline — from the month-25 marketing-mix shift, MX acquisition tilts ~50% to a telco bundle + affiliate. Low-quality-channel cohorts churn +4–5 pp while young; MX gross churn drifts up from ~month 28 (4.7% → 5.9%) <i>before</i> the visible spike, then F1 accelerates it. OXOX/carrier involuntary churn worsens too.	Pre-existing drag (MX)	Acquisition Quality & Channels (channel × market), Market Context

US is the control: card-only (no dunning tail), no price increase, little bundle, little F1 → US gross churn stays ~4% throughout. No single report page shows all three factors; the diagnosis needs Content&Engagement + Pricing&Plans + AcquisitionQuality + MarketContext read together.

3.2 Calibration

Calibrated against the world truth over ~10 passes at POP_SCALE 0.05–0.25 (churn rates are scale-invariant). Final full run (POP_SCALE=1.0):

Metric	Result	Target
Pre-shift blended gross monthly churn (idx 24–32)	4.11 %	3.7–4.4 %
Peak blended (idx 33–35)	6.82 %	~7 % ("nearly doubled")
Peak / pre ratio	1.66×	~1.7–2.0×
BR churn idx 32 → 33	4.0 % → 7.0 % (step)	acute
MX churn idx 26 → 32 → 34	5.3 % → 5.9 % → 8.3 % (ramp)	gradual then accelerating
US churn idx 26 → 34	4.3 % → 4.4 % (flat)	control
Cohort retention M1 / M6 / M12	89 % / 74 % / 64 %	M12 a touch above the doc-01 band
Active subscribers at window end	~66.8 k	~70 k
DQ checks	6 hard PASS, 0 soft warnings	all pass

3.3 Problems hit (for the playbook guardrails)

- **Staging was ~45 min** at full scale: row-wise `.map(parse_date_any)` / `.map(parse_money)` / per-row `norm()` on the 2–4 M-row Ketch tables and the 17 M-row PlayLog/Trackpad loop. Rewrote the hot paths as **vectorised** `parse_date_series` / `parse_money_series` / `norm_series` (Series `.str + .map(dict)`), and PagStream `meta` JSON via vectorised regex extract instead of `json.loads` per row. Staging dropped to **141 s**.
- `fact_subscription_month` folded from the *shuffled* ledger initially lost ~15 % of churn events because two events in the same month were ordered by their random timestamps. Fixed by sorting on an explicit event-type priority before replay.
- Recording bug: churned-this-month rows were skipped (subscriber already `status=3`), so churn never landed in `fact_subscription_month`. Now emits active + paused + churned-now.
- Content amortisation was ~10× too high (P&L ≈ −\$10 M/month on ~\$1.1 M MRR). Recalibrated `content.py` cost draw so amortisation lands near ~50–60 % of subscription revenue.
- Raw generation ≈ 19 min at full scale (~17 M PlayLog rows across 36 monthly Parquet files).

Phase 4 — Data documentation

- `docs/03_data_dictionary.md` — **auto-generated** by `etl/gen_data_dictionary.py` from `data/curated/`: 24 tables, grain guess + row count + every column with dtype and sample values.
- `docs/04_etl_pipeline.md` — stage-by-stage narrative: the generative world model (`gen/world.py` + `content.py` + `economics.py`), raw messiness per source, the staging fix classes and the vectorisation that took staging 45 min → 2.5 min, the curated star build (ledger fold-forward with event-priority ordering, sentinels, USD everywhere), and the DQ gate. The churn-shift mechanism is referenced as `config.py §9` without enumerating the factors (kept for the Phase-10 diagnosis).
- `docs/05_data_model.md` — bus matrix (13 facts × 11 dims), per-fact grain + measures + degenerate dims + row counts, dimension attribute list, the relationship list (single direction, `subscriber_key` as the hub, one active date role, `dim_price_history` and `fact_fx_rate` unrelated/DAX-resolved), and the Total / Comparable Base / Expansion cut.
- `docs/08_data_quality.md` — Q01–Q15 each mapped injected(count) → handled → asserted with real numbers from the shipped run; the 6 hard + 7 soft check results; and 8 accepted limitations (sampled acquisition channel, schedule-based billing amount, partial last month, measured-not-corrected snapshot drift, M12 retention ~64 % slightly high, thin ramping gross margin, no external data).

Phase 5 — Visual identity

- `assets/brand.py` — the palette + type single source of truth (imported by `gen_logo.py` now and `gen_report_visuals.py` in Phase 8). Brand violet `#4B2E83`, one accent amber `#E6A23C`, one comparison steel `#5B7FB0`, muted lavender-grey ground `#F2F1F7`, quiet `#E7E5EF` gridlines,

good/bad #2E7D5B / #C0453B (churn = lower-is-better, so deltas invert).

- `assets/gen_logo.py` (Pillow) — emits the rounded-tile **play + amber-plus mark** (on-light / on-dark), the **Voxa+ wordmark**, and **7 full-page background PNGs**: violet band + wordmark + a quiet "Subscriber Churn Diagnosis" label, amber hairline, then the page title + business question on the light body. Also writes `_page_heights.json` (Executive 900, rest 720).
- `powerbi/theme/theme.json` — `visualStyles` for ~20 visual types + `page`; **passes pbir theme validate**. Soft drop shadow on charts only; horizontal gridlines only; cards transparent (a white rounded frame shape supplies the surface); slicers headerless; waterfall sentiment pinned to the KPI-delta colours.
- `docs/07_visual_identity.md` — the written identity spec + the signature-element recipe repeated on every page (band → title → Currency/Market slicers → Analysis-Period slicer → 6 KPI cards → two 3-30-300 content rows).

Phase 6 — Semantic model as code

- Added `fact_engagement_device_month` to the curated star (market x device family x month, ~500 rows, with `vsf_rate` / `playback_error_rate`) and a `date_key` to `fact_billing_attempt`, then re-ran `--from 03` (dq still 6/6 + 0 warnings). This keeps the 13.5 M-row `fact_viewing_daily` **out of the semantic model** — it stays analyst-only for Phase 10 — while the report still gets a device-level QoE cut. `fact_engagement_device_month` cleanly shows F1: CTV video-start-failure 1.1 % -> 2.6 % (idx 32) -> 3.4 %, non-CTV flat.
- `powerbi/gen_semantic_model.py` generates the whole `.pbip` from the curated Parquet schemas: `VoxaChurnDiagnosis.pbip` + `.SemanticModel/` (TMDL: `database`, `model` (auto date/time **off**), `expressions` (`pDataFolder` parameter), `relationships`, `tables/*.tmdl` — each with a `Parquet.Document` import partition) + a `.Report/` shell. **24 imported tables + 5 helper tables** (`Reporting Currency`, `P&L Line`, `Subscriber Bridge`, `Funnel Stage`, `_Measures`) built from M `#table` literals. **34 relationships**, `dim_subscriber` as the hub, direct `dim_market` links only for subscriber-less aggregate facts, `dim_content[release_date_key]` inactive.
- Problems hit: `pandas Path.write_text` on Windows emitted CRLF -> routed every TMDL/JSON write through an LF helper; `datetime64[ms]` / `datetime64[us, UTC]` weren't in the string dtype map -> switched to `pandas.api.types` checks so date columns type as `dateTime`, not `string`.
- `powerbi/README.md` documents the one-time Desktop pass (point `pDataFolder`, Refresh, mark `dim_date`). Structural self-check passes: 29 tables, all relationship column refs and all partition file refs resolve.

Phase 7 — DAX measures

- Two small curated additions first (re-ran `--from 03`, dq still 6/6 + 0): `dim_date[abs_month]` (`year*12 + month_num`, a linear index for time intelligence) and `dim_subscriber[is_incentivised]` (sampled from the channel's promo-code rate — same caveat as the sampled channel).
- `etl/pbi_measures.py` — `MEASURE_LIBRARY`: **107 measures in 15 display folders** (`Currency plumbing` · `Subscribers` · `Churn` · `Cohorts` · `Engagement` · `App Quality` · `Payments` · `Revenue` · `Unit Economics` · `Time Intelligence` · `Comparable Base` · `Targets` · `Acquisition` · `Support` · `Narrative`). Design choices:
- **Time intelligence with no marked date table**: `PY = abs_month - 12` via `TREATAS(SELECTCOLUMNS(VALUEs(dim_date[abs_month]), ... , -12), dim_date[abs_month])`; YTD / MAT / MoM all use `abs_month` arithmetic. No `SAMEPERIODLASTYEAR`.
- **Reporting currency**: [`FX Rate to Reporting`] (`USD`→selected, date-context aware) is multiplied into every leaf monetary measure; [`Currency Symbol`] for labels/narrative.
- **Gross Monthly Churn %** = mean of the monthly `churned / start-of-month active` ratios (matches the `dq_checks.py` definition). [`Pre-Shift Baseline Churn %`] fixes the window-month 25-33 band; [`Churn vs Baseline (x)`] gives the "nearly doubled" multiple.
- **Cohort retention** via [`Retention %`] (`cohort_month × tenure_months` matrix) plus explicit `M1/M3/M6/M12 Retention %`.
- **Targets** one ... vs Target % per KPI; churn / CAC flagged lower-is-better (negative = good) in the description and inverted on the KPI cards in Phase 8.
- [**Exec Insight**] — a dynamic narrative text measure that reports the churn level vs baseline, the worst market and net adds, and updates with the slicers (replaces the Smart Narrative visual).
- `powerbi/gen_semantic_model.py` now injects the library into `_Measures.tmdl` (TMDL `measure` blocks with `/// description`, `formatString`, `displayFolder`, `lineageTag`).
- `etl/gen_dax_doc.py` -> `docs/06_dax_measures.md` (auto-generated, DAX in fenced blocks).
- Self-check: all 107 measures present and quoted; every [`Measure`] reference resolves to a defined measure or a real column. DAX semantics get their first real check on the Phase-8 Desktop open.

Phase 8 — Report build

- **pbir reads the code-generated TMDL model from disk** (thick `byPath` report), so the whole report can be built and field-validated here without Power BI Desktop. Flow: `pbir new report --thick --no-title -> pbir report rebind --local ../<name>.SemanticModel -> per page pbir add visual --from-json <bundle> -> pbir visuals sort / pbir visuals cf -> report-level dim_date[in_analysis_window] filter -> pbir validate --all`.
- `powerbi/gen_report_visuals.py` -> `powerbi/_report_layout.json`: 7 pages, the signature furniture (bg PNG, 3 slicers, white KPI frame + 6 value cards + delta cards), two content rows each. `powerbi/_drive_pbir.py` turns the manifest into `pbir` calls (retry-on-lock, `PBIR_ALLOW_OVERLAP=1` for the layered KPI band). `powerbi/build_report.sh` orchestrates (abort if Desktop running, regen `assets/model/manifest`, drive pbir).
- **Result: 7 pages, 113 visuals, 239 fields bound, pbir validate --all PASSES** (0 errors). 188 warnings remain, all from the deliberate dense-KPI design: `VISUAL_OVERLAP` (cards on the frame), `VISUAL_UNDERSIZED` (compact KPI tiles), `TOO_MANY_VISUALS` (~20 on 4 pages) — the kind `pbir validate` flags but only a Desktop screenshot can actually judge.

- Problems hit: `--from-json` only accepts keys `visual_type/name/title/x/y/width/height/` fields (no `z`, no `objects`); role names are `Category / Y / Series` (charts), `Rows / Columns / Values` (matrix), `Values` (card/slicer/table) — **LineChart has no Series role**, so series breakdowns use `clusteredColumnChart`; subprocess needed `encoding="utf-8"` for pbir's box-drawing output; `definition.pbism` needed a `$schema`; 720px pages needed the content rows remapped (tuned for the 900px exec page).
- `pbir theme apply-template` only takes registered template *names*; our authored `theme/theme.json` is applied in Desktop (View > Themes > Browse) on first open.
- `docs/09_report_structure.md` — the page-by-page spec + the first-open checklist (`pDataFolder`, theme, mark `dim_date`).
- Next: the user opens `powerbi/VoxaChurnDiagnosis.pbip` in Desktop and we iterate on screenshots** (layout polish, KPI-card formatting, per-visual data-label toggles, the chronological sort verification).

Phase 8 — Desktop iteration (round 1)

Opened the `.pbip` in Desktop and drove fixes through the `pbir desktop` bridge (`PBIR_ADOMD_DIR="C:\Program Files\Power BI Report Builder"` for the live-model queries):

- `compatibilityLevel 1550 → 1600` — Desktop auto-upgrades on open and refuses a downgrade on the next open.
- Friendly tab names** via `pbir set " ... Page.displayName"` (folder / internal id kept): Executive Summary · Retention & Cohorts · Content & Engagement · Pricing & Plans · Acquisition Quality · CX, Billing & App · Market Context.
- Removed the default Title.Visual textboxes** `pbir add page` injects (the background PNG carries the title).
- Chronological x-axis:** `dim_date[year_month]` and `dim_subscriber[cohort_month]` given `sortByColumn` in the TMDL; every time-series visual then `pbir visuals sort -ed` by its Category ascending (new charts default to sort-by-first-measure-descending).
- KPI measures fixed** — the headline cards were summing/averaging levels across all 36 months:
- `Paid Active Subscribers` → mean of the monthly active counts (was `DISTINCTCOUNT` over the whole window = 107 k).
- New `... (Latest)` wrappers anchor to the last in-window month (`window_month_idx`, not `MAX(abs_month)` which hit the padded `dim_date` end at Dec 2027 and returned blank). Exec + Market-Context KPI cards use them; Exec Insight too.
- `LTV` guarded with `MAX(0, ...)`.
- Verified against the live model:** `[Exec Insight] = "Gross monthly churn is 7.0%. That is about 1.7x the pre-shift baseline of 4.1% and the highest of the three markets is Mexico (8.3%). Net adds in the latest month: 205."` — the diagnosis reads correctly.

Open polish items (left for the next Desktop round; the model + data are verified correct): apply `powerbi/theme/theme.json` (View > Themes > Browse); a **Home > Refresh** to clear Desktop's stale visual cache; the KPI frame `shape fill`; a (blank) month on two exec charts; the compact KPI value/delta cards (`VISUAL_UNDERSIZED` is by design).

Phase 10 — Data analysis (dual-track)

- `data_analysis/run_analysis.py` — **10 questions, each computed twice** (DuckDB SQL over the curated Parquet, and pandas over the same files) with `lib/parity.assert_parity` asserting the two agree before any number is used. `python data_analysis/run_analysis.py → sql/qNN.sql (17), results/qNN*.csv (17), parity/parity_report.md (14/14 checks pass, all exact)`.
- Bugs found and fixed while building it: `LAG()` evaluated after `WHERE` in the same SELECT (moved the window into its own CTE); a `LEFT JOIN` on a non-distinct `m3` set fanning out the aggregate (`SELECT DISTINCT`); orphan-billing rows with null `market` inflating a SQL `GROUP BY` (filter `market IS NOT NULL`); a per-row `.loc` lookup in Q4 replaced with a vectorised shifted merge.
- The diagnosis** (`findings/F01 – F10`, `deliverables/deep_dive_00_index.md`): churn 4.1% → 6.8% (1.66×, 90% voluntary, not receding); **three overlapping factors**, none sufficient alone — 1. **F05** connected-TV app-v3 regression: CTV video-start-failure 1.1% → 3.0% at window-month 32; CTV-heavy churn 4.1% → **8.4%** vs 4.1% → 5.6% for the rest. 2. **F06** Brazil price increase: **~0 pp** on engaged Std/Prem, **+18 pp** on already-disengaged — the amplifier, Brazil-only. 3. **F07** Mexico cohort-quality drag: post-mid-2025 Mexican cohorts retain **–8–9 pp** at month 3 (vs **–2–3 pp** elsewhere) — the pre-existing drain.
- F08** rules payments out (failure / dunning / involuntary all flat).
- F09:** implied lifetime 24 → 15 months (**–40% LTV**); ~5,400 excess churned in 3 months (~\$0.5–0.6 M lifetime revenue).
- F10** decomposition: in Brazil 85% of peak voluntary churn carries F1 or F2; in Mexico ~64% carries F1 or F3; US "39% none" is just its unchanged baseline (the control).
- `deliverables/board_summary.html` — one-page, decision-first, Voxa+ identity.

Phase 11 — Portfolio assets

- `portfolio/gallery/` — **5 gallery images**, rendered to 3200×1640 PNG with headless Chrome. Four are self-contained HTML (Voxa+ identity, no external assets): medallion pipeline, semantic-model bus matrix, DAX library, report-structure map. Image 1 frames a live Executive Summary screenshot (`powerbi/_shots/round2/`) inside the same HTML template.

Phase 12 — Docs bundle & delivery

- `docs/build_docs.pdf.py` — collates the docs (markdown → one styled HTML: navy headers, amber rule, navy table headers + zebra rows, amber callout quotes, dark mono code blocks, A4 with controlled page breaks; mermaid fences flattened to plain code) → Chrome `--headless=new --no-pdf-header-footer --print-to-pdf`.
- `docs/Voxa+ Documentation.pdf` — full set (README + 01–09 + deep-dive + walkthrough), ~2.9 MB.
- `docs/Voxa+ Summary.pdf` — condensed (context + DQ + diagnosis), **0.7 MB (< 2 MB)**.

- **Trimmed the curated CSVs:** Parquet is the source of truth; `03_build_curated._save` now writes a CSV mirror only for tables ≤ 200 k rows. Removed the 4 large fact CSVs (~360 MB); `data/curated/` went 689 MB $\rightarrow$ 344 MB.
- Finalised `README.md` (the diagnosis up top, a layout map, a reproduce block).
- **Sync:** the delivery folder copy (e.g. Google Drive) is the last manual step — the project lives under `PBI Pessoal/voxaplus-churn-diagnosis/`.

Project complete (Phases 0–12). Known follow-up: a final Power BI report polish round via the `pbir desktop` loop — **Home > Refresh** to clear the stale visual cache the current gallery image 1 and `_shots/` captures still show (a couple of ghosted KPI-card labels).

Phase 8 — Desktop iteration (round 2, file-level)

Applied without Desktop (the bridge was unavailable):

- **Theme applied.** `pbir theme serialize powerbi/theme/theme.json -o _vox.Theme` $\rightarrow$ `pbir theme build _vox.Theme -o VoxaChurnDiagnosis.Report -f --clean` bakes "Voxa Plus Diagnosis" into `report.json` (27 visual types) — replaces the default sqlbi theme that was giving the KPI band its blue look. `_drive_pbir.py` now does this on every build.
- **KPI frame shapes** $\rightarrow$ white fill, no drop shadow, on all 7 pages (`pbir set "**/*kpiframe*.Visual.fill.{show,fillColor}"`).
- **Narrative cards** (`exec_insight` , `insight_7`) $\rightarrow$ 10 pt text + word wrap so the dynamic `[Exec Insight]` paragraph fits.
- `pbir validate --all` passes (178 warnings, all the by-design KPI overlap / undersized).

Still needs a Desktop pass by the user: **Home > Refresh** (the open instance still holds a stale visual cache from before the measure fixes), then re-capture the Executive Summary screenshot gallery image 1 frames.